

6

Force due to Magnetic Field

The Sun and other stars eject many high-energy charged particles every second. These charged particles are what make up cosmic rays. Many of these particles are deflected away from the Earth by the Earth's magnetic field. Some are led to the region around the magnetic poles of the Earth. They enter the Earth's atmosphere and collide with air molecules to emit light of different colours, forming the beautiful aurora.

In this chapter, we will learn about

- the magnetic force acting on a current-carrying conductor
- the working principle of a motor
- the magnetic force acting on a moving charge

6.1 Magnetic force on current-carrying conductor

1 Direction of magnetic force

a Fleming's left-hand rule

DSE goal

Examine existence of force on current-carrying conductor in magnetic field and determine the relative directions of force, field and current.

DSE exam

13(1A)Q26, 28, 15(1B)Q9(a), (c), 20(1A)Q26

In Chapter 5, we learned that a current-carrying conductor produces a magnetic field. What happens if the current-carrying conductor is put inside an external magnetic field?

Here, an external magnetic field means a magnetic field not produced by the current passing through the conductor.

Video

→ This video demonstrates Expt 6a.

Flipped classroom

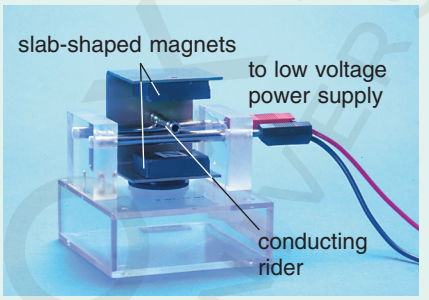
Watch a video and answer the questions.

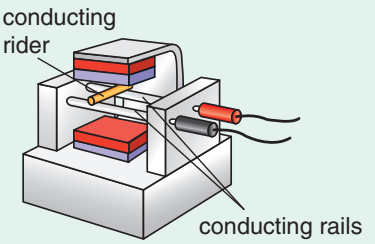
Experiment 6a Magnetic force on a current-carrying conductor

1 Set up the Fleming's apparatus. Switch on the power supply and observe any motion of the rider.

2 Repeat by reversing the current direction.

3 Repeat by reversing the poles of the magnets.





Discussion

How are the directions of the force, the current and the magnetic field related?

When a current I passes through the rider in a magnetic field B, the rider moves. This shows that a magnetic force F acts on the rider (Fig 6.1a). If the direction of **either** (but not both) I or B is reversed, the direction of F is also reversed.

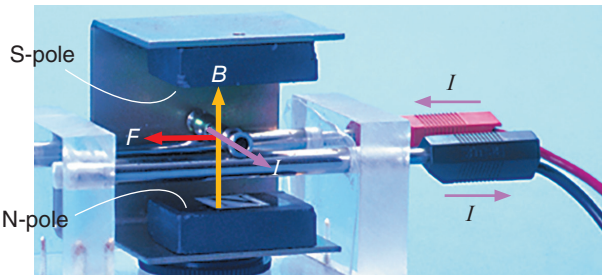
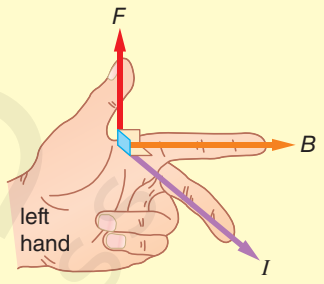


Fig 6.1a *A current-carrying conductor (rider) moves in a magnetic field.*

The directions of the magnetic force, the current and the magnetic field are related by **Fleming's left-hand rule**.

Hold the thumb and the first two fingers of the left hand at right angles. The thumb gives the direction of the magnetic force *F*, if the first finger points in the same direction as the external magnetic field *B*, and the second finger in the same direction as the current *I*.

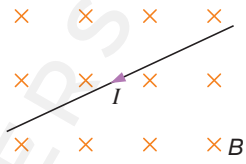


Remind Ss that this is a rule (not a physics principle) that helps them remember the directions of the force, magnetic field and current.

May ask Ss what happens to the direction of *F* if both *I* and *B* are reversed.

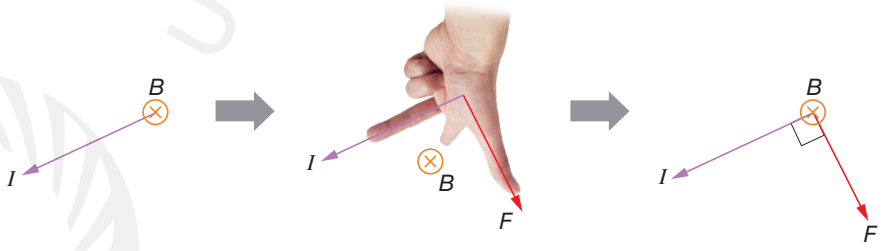
Example 1 Applying Fleming's left-hand rule

A wire carrying current *I* is situated in a uniform magnetic field *B* pointing into the page as shown. Find the direction of the magnetic force *F* acting on the wire.

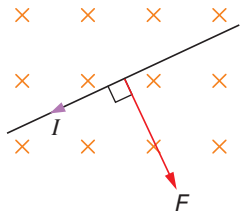


Solution

Apply Fleming's left-hand rule as follows.



The direction of the magnetic force *F* is as shown.



Know more

John Ambrose Fleming (1849–1945)

Sir *John Ambrose Fleming* was an English electrical engineer and physicist. He devised the left-hand rule and the right-hand rule (see Chapter 7). He is also renowned for the invention of the thermionic valve (vacuum tube), which is regarded as the birth of electronic devices.



Moving-coil loudspeakers

Do you know how a moving-coil loudspeaker works?

▸ p.264

Battery train

You can make a battery train using a copper wire, a battery and some magnets.



Video

▸ p.264

→ This video demonstrates the battery train.

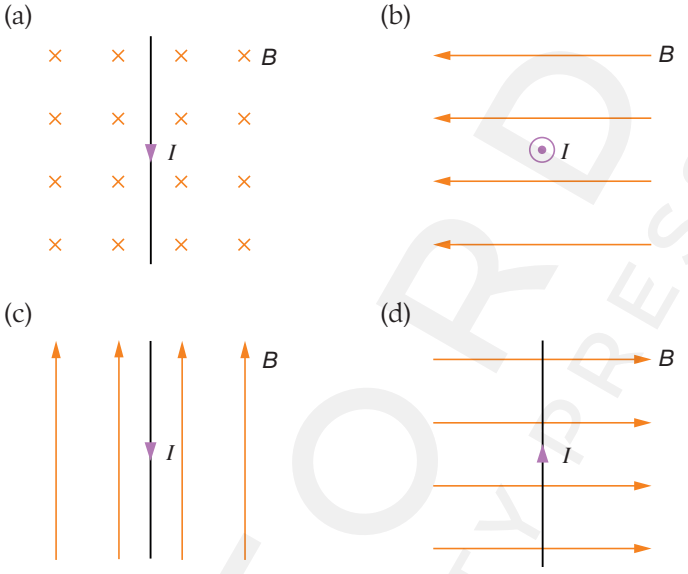
Fleming's left-hand rule 弗林明左手定則　John Ambrose Fleming 弗林明

Google Forms Microsoft Forms

Quick check 1

→ E-version in OUP Exercise Platform is also available.

1 Find the direction of the magnetic force acting on the current-carrying conductor.



b Resultant field and magnetic force

Remind Ss that this is a method (not a physics principle) that helps them remember the direction of the force.

When a current-carrying straight wire is placed inside a uniform magnetic field (Fig 6.1b), besides using Fleming's left-hand rule, the direction of the magnetic force on the wire can also be found by considering the resulting magnetic field around the wire (Fig 6.1c).

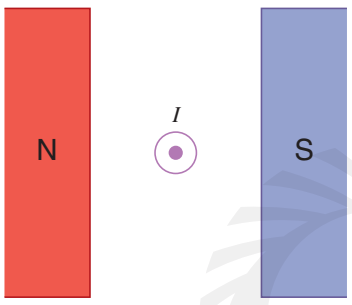


Fig 6.1b *A current-carrying straight wire inside a uniform magnetic field.*

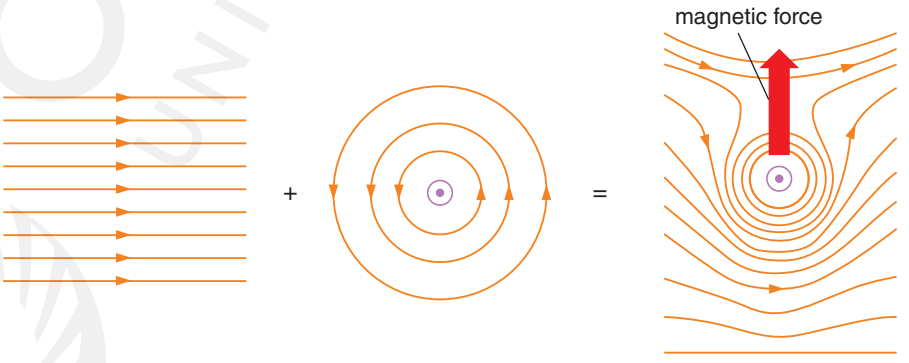
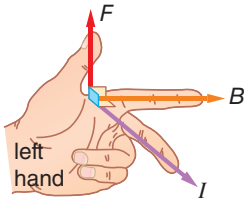


Fig 6.1c *The resultant magnetic field around a current-carrying straight wire placed in a uniform magnetic field.*

This agrees with Fleming's left-hand rule, with *B* being the external uniform field.



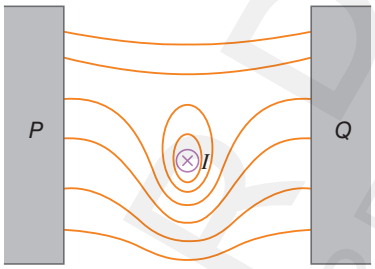
In the resultant field, the field lines are denser below the wire, where the two original fields reinforce each other. The resultant field lines are less dense above the wire, where the two original fields cancel each other.

This resultant field is called a *catapult field* because the field pattern looks like a catapult. The wire experiences a magnetic force in a direction similar to an object being propelled by a catapult.

catapult field 弓形磁場

Example 2 | Catapult field

A straight wire carrying current I pointing into the page is placed in the uniform magnetic field between magnetic poles P and Q. The figure shows the resultant field pattern.



(a) What is the direction of the magnetic force acting on the wire?

(b) Is the magnetic field produced by the current clockwise or anticlockwise?

(c) Is P an N-pole or S-pole?

Solution

(a) Upwards

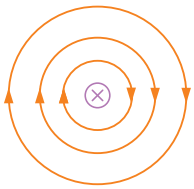
(b) Clockwise

(c) S-pole

Tip

By Fleming's left-hand rule, the external field (due to magnetic poles) points to the left. Therefore, Q is an N-pole and P is an S-pole.

Alternatively, B due to I:



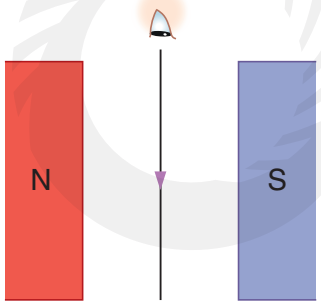
In the resultant field, the field lines are denser below the wire. Therefore, the fields due to I and the magnetic poles point in the same direction (towards left) below the wire.

Google Forms Microsoft Forms

Quick check 2

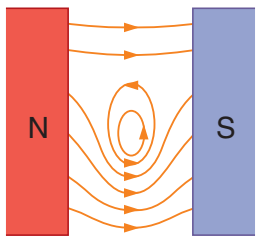
→ E-version in OUP Exercise Platform is also available.

1 In the figure below, a long straight current-carrying wire is placed between two slab-shaped magnets which produce a uniform magnetic field.

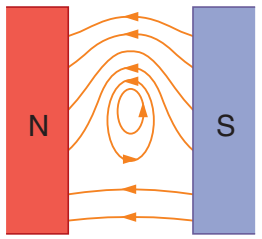


Which of the following correctly shows the resultant magnetic field pattern between the magnets as observed from above?

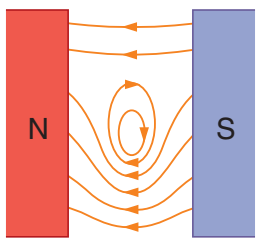
A



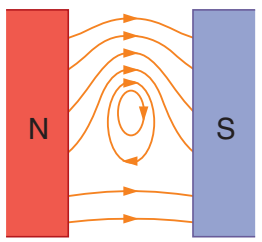
B



C



D



DSE goal

Determine factors affecting force on straight current-carrying wire in magnetic field and represent the force by $F = BIl \sin\theta$.

DSE exam

15(1B)Q9(b), 23(1B)Q8

2 Magnitude of magnetic force

We have learned how to determine the direction of the magnetic force acting on a straight current-carrying wire in a magnetic field. Now, we will study the factors affecting the magnitude of the magnetic force. We will use a current balance to find out the answer (Fig 6.1d).

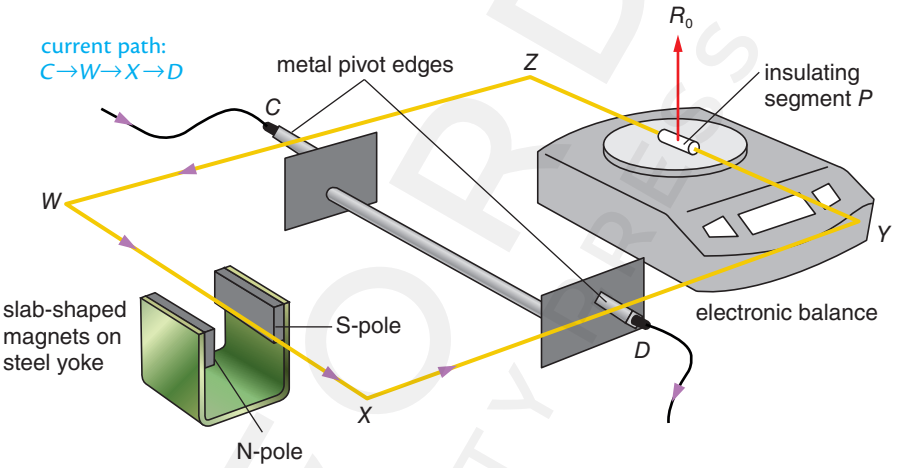


Fig 6.1d *A current balance.*

CW should be shorter than *CZ* to keep the frame in contact with the balance.

In the current balance, a copper wire frame *WXYZ* rests on metal pivot edges *C* and *D*. Two slab-shaped magnets set up a uniform magnetic field perpendicular to *WX*. Insulating segment *P* rests on the electron balance and the balance reading is set to zero.

The normal reaction acting on *P* by the balance is R_0. Since the frame is at rest, the moment due to the weight of the frame and that due to R_0 balance each other. Let us assume that *P* just touches the balance and $R_0 = 0$ for simplicity.

C and *D* are then connected to a d.c. power supply. A current will pass through segment *CWXD* but not *CZYD* because the insulating segment *P* breaks the circuit at *ZY* (Fig 6.1e).

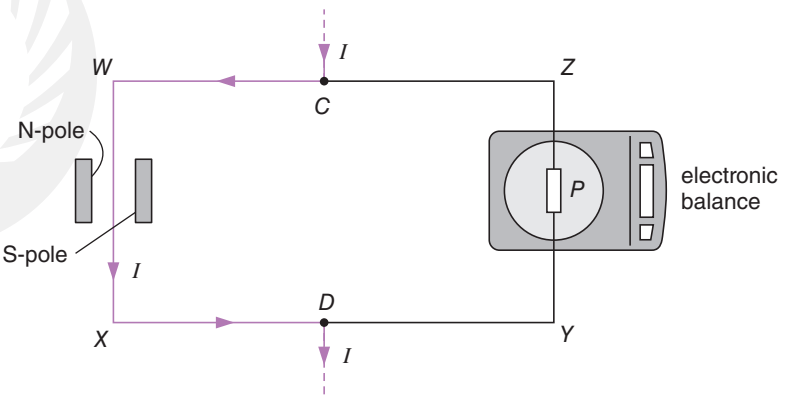


Fig 6.1e *A current passing through segment CWXD but not CZYD.*

Suppose $R_0 \neq 0$.
Take moment about the pivot.
Let the moment due to the weight of the frame be τ_W.
Before passing a current through the frame,
clockwise moment
= anticlockwise moment
$\tau_W = R_0(CZ)$
Suppose the reading of the balance is *x* kg when a current passes through *WX*.
$R = x(g) + R_0$
Clockwise moment
= anticlockwise moment
$F(CW) + \tau_W = R(CZ)$
$F(CW) + R_0(CZ) = (xg + R_0)(CZ)$
$F(CW) = xg(CZ)$
$\therefore$ *F* can still be found from *x*.

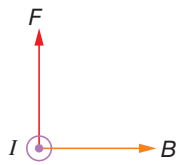
When the current flows from W to X, by Fleming's left-hand rule, an upward magnetic force F acts on WX (Fig 6.1f). The magnitude of F can be worked out from the balance reading, which shows the size of the normal reaction R acting on the frame by the balance (Fig 6.1g).

Cross-link

You learned the relationship between the balance reading and normal reaction in Book 2 Ch 3. You also learned moment in Book 2 Ch 5.

Tip

When viewed from X to W,



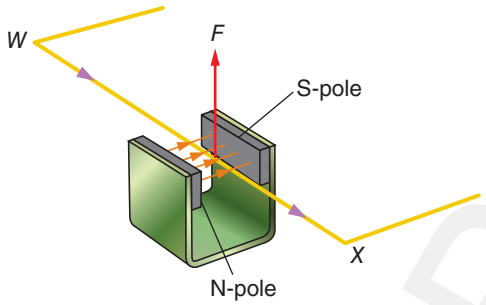


Fig 6.1f The magnetic force acting on segment WX.

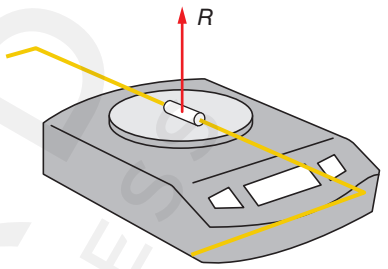
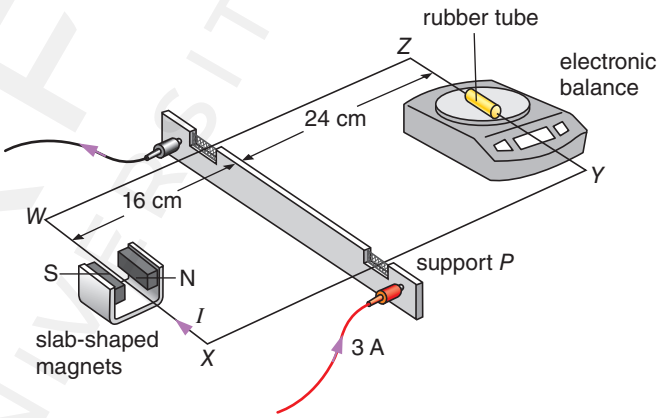


Fig 6.1g The normal reaction R acting on the frame by the balance.

Example 3 Finding magnetic force using a current balance

A student sets up a current balance as shown. When there is no current, the frame is at rest. Assume the rubber tube just touches the electronic balance and the reading of the balance is zero.



Now, a current I flows from X to W. The frame remains at rest. The balance reading is 0.15 g.

(a) Determine the direction of magnetic force F acting on WX.

(b) Find the normal reaction R acting on YZ by the balance.

(c) Find F.

(d) How will the balance reading change if F doubles?

Solution

(a) Upwards

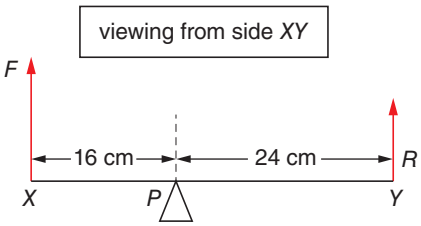
(b) $R = 0.15 \times 10^{-3} \times 9.81 = 1.47 \times 10^{-3}$ N

(c) Since the frame remains at rest, by taking moment about P,

$F \times 0.16 = R \times 0.24$

$F \times 0.16 = (1.47 \times 10^{-3}) \times 0.24$

$F = 2.21 \times 10^{-3}$ N



(d) Since $F \propto R \propto$ balance reading,

the balance reading will double if F doubles.

Exercise 6.1 Q4 (p.233)

Using the current balance, we can carry out the following experiment to find out the factors affecting the magnetic force.

Simulation Video

→ This simulation demonstrates the factors affecting the magnetic force on a straight wire. It is available on OUP web in PC version (.exe) only.

→ This video demonstrates Expt 6b.

Experiment 6b Factors affecting magnetic force on a straight wire

1 Set up the current balance and set the balance reading to zero (Fig a).

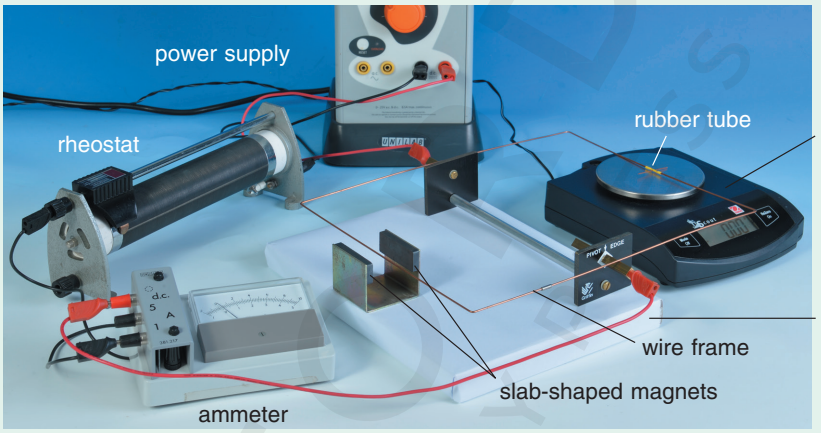


Fig a

✎ To have a more accurate result, align the current-carrying arm along the N-S direction to minimize the effect of the Earth's magnetic field.
Place the set-up far from the disturbance of wind.

2 Switch on the power supply. Adjust the rheostat to pass different sizes of current through the wire frame and record the balance readings.

3 Fix the current and add more pairs of slab-shaped magnets one by one next to the original pair (Fig b). Make sure the magnetic fields produced by the magnets are in the same direction. This increases the length of the wire frame inside the magnetic field. Record the balance readings.

4 Fix the current and stack more pairs of slab-shaped magnets one by one to strengthen the magnetic field (Fig c). Record the balance readings.

✎ In step 4, the results are approximately linear if new, identical and thin slab-shaped magnets are used.

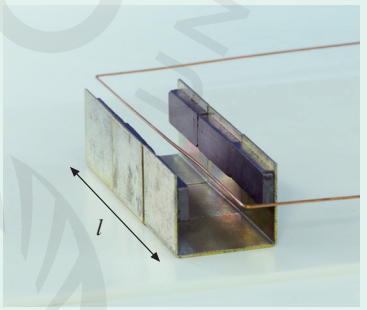


Fig b

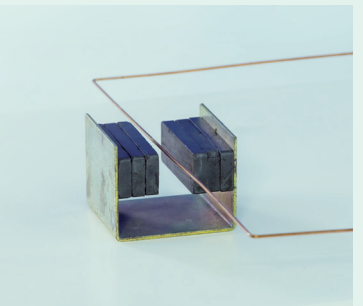


Fig c

Discussion

1 How does the magnetic force F change in steps 2–4?

2 Suggest a safety precaution in this experiment.

The rheostat should be set to its maximum value at the beginning. / Switch off the power supply after taking each reading.

We can see that the magnetic force F acting on a straight current-carrying wire depends on the current I passing through the wire, the length l of the wire **inside the magnetic field**, and the magnetic field B.

In fact, F, I, l and B are related as follows (Fig 6.1h).

$F \propto I$ (constant B and l)

$F \propto l$ (constant B and I)

$F \propto B$ (constant I and l)

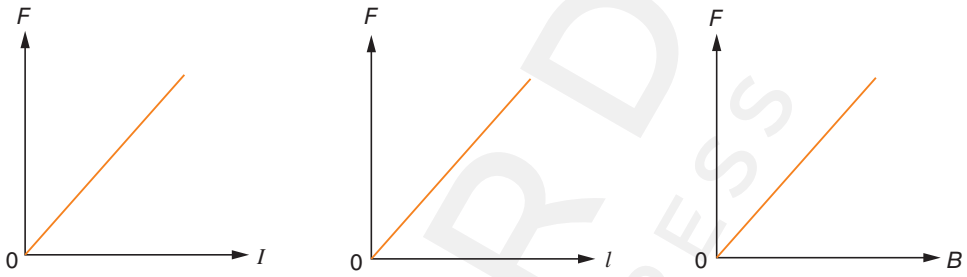


Fig 6.1h *Graphs showing $F \propto I$, $F \propto l$ and $F \propto B$.*

Combining the three relationships, we have

$F \propto BIl$ or $F = k \times BIl$, where k is a constant.

Cross-link

In Book 2 Ch 3, Newton's 2nd law can be written as $F = kma$.

The unit newton (N) is defined to make $k = 1$.

▶ Scientists define the unit tesla (T) in a way that $k = 1$. Therefore,

$F = BIl$ (with $B \perp I$)

$$B = \frac{F}{Il}$$

A straight wire carrying a current of 1 A which is perpendicular to a magnetic field of 1 T will experience a magnetic force of 1 N per unit length (1 m).

▶ When $F = 1$ N, $I = 1$ A and $l = 1$ m, $B = 1$ T. This defines the tesla.

$$\mathbf{1\ T = 1\ N\ A^{-1}\ m^{-1}}$$

If the direction of the current I and the magnetic field B makes an angle θ, only the perpendicular component of B contributes to the magnetic force F (Fig 6.1i).

In Experiment 6b, it is not necessary to align the wire perpendicular to the magnetic field. Ts may ask Ss to do Revision 6 Q17 (p.260) to prove this.

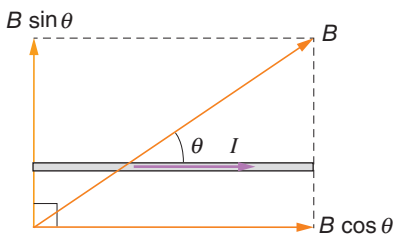
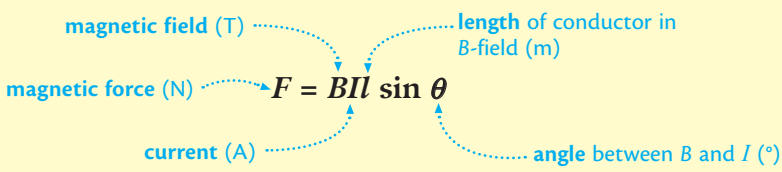


Fig 6.1i *Resolving magnetic field B.*

Then, the magnetic force acting on a straight current-carrying wire is



Note the following:

F is always perpendicular to *B* and *I*.

1 The direction of *F* can still be found by using Fleming's left-hand rule. The first finger points along the component of *B* perpendicular to *I*. (Fig 6.1j).

2 *F* = 0 when *B* and *I* are parallel to each other, i.e. $\theta = 0°$ ($\sin\theta = 0$) (Fig 6.1k).

3 *F* attains the maximum when *B* and *I* are perpendicular to each other, i.e. $\theta = 90°$ ($\sin\theta = 1$) (Fig 6.1l).

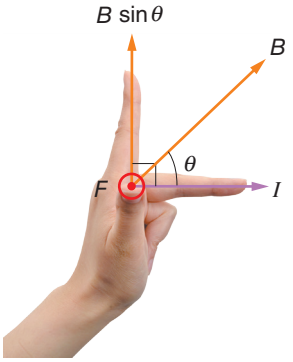


Fig 6.1j *Applying Fleming's left-hand rule.*

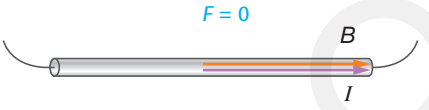


Fig 6.1k *B parallel to I.*

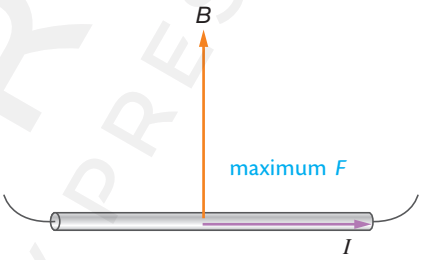
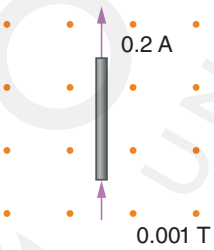


Fig 6.1l *B perpendicular to I.*

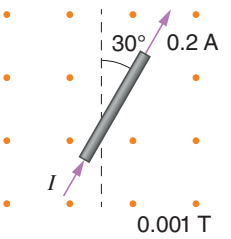
Example 4 Calculating magnetic force on a rod

A conducting rod of length 0.15 m is placed in a uniform magnetic field of 0.001 T. Given that a current of 0.2 A passes through the rod, find the magnitude *F* of the magnetic force acting on the rod in each of the following cases.

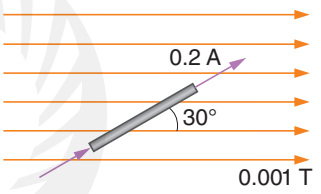
(a)



(b)



(c)



Solution

(a) $F = BIl = 0.001 \times 0.2 \times 0.15 = 3 \times 10^{-5}$ N

(b) $F = 3 \times 10^{-5}$ N

(c) $F = BIl \sin\theta = 3 \times 10^{-5} \times \sin 30° = 1.5 \times 10^{-5}$ N

Tip

The rod (current) is still perpendicular to the magnetic field, so the answer is the same as (a).

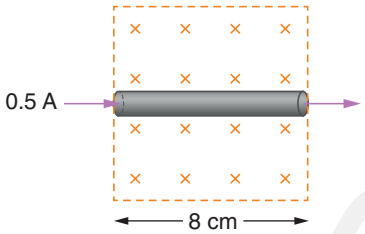
Google Forms Microsoft Forms

Quick check 3

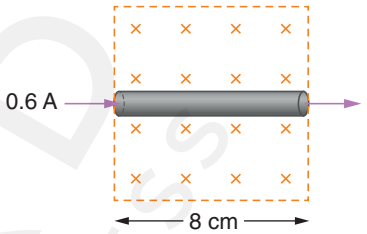
→ E-version in OUP Exercise Platform is also available.

1 In each of the following cases, current-carrying rods are placed in the same uniform magnetic field which covers an area of 8 cm × 8 cm. Which rod experiences the largest magnetic force?

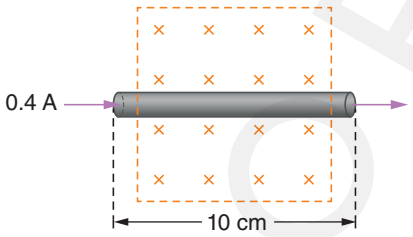
A



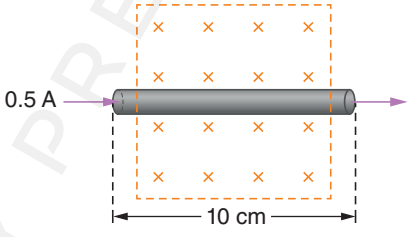
B



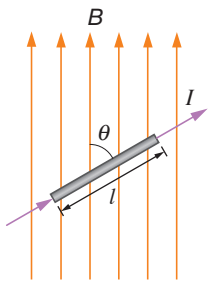
C



D



2 When a conductor of length l carrying a current I is placed in a uniform magnetic field B, it experiences a magnetic force of magnitude F. Complete the following table.

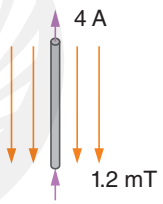


	B	I	l	θ	F
(a)	1.5 mT	12 A	0.2 m	45°	
(b)	0.8 mT		0.12 m	60°	1.4×10^{-3} N
(c)		8 A	0.25 m	30°	2.0×10^{-3} N

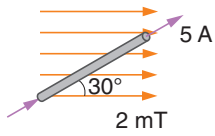
Learn by practice 1

1 Find the magnitude and the direction of the magnetic force acting on the conductor in each of the following cases. The length of the conductor within the magnetic field is 0.2 m.

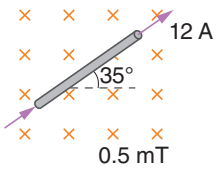
(a)



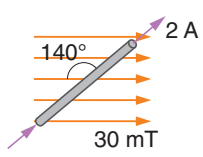
(b)



(c)



(d)



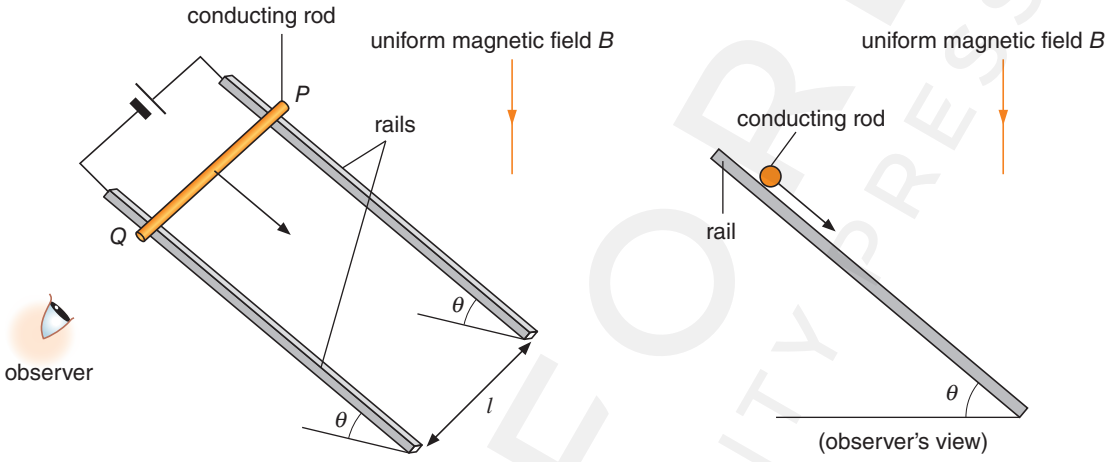
☆ HKDSE 2013 Paper 1A Q28

→ This video demonstrates how this question is solved.

DSE practice 1 | Conductor rolling down inclined rails

Video

A conducting rod PQ of mass m slides down along two parallel conducting rails inclined at θ to the horizontal. The separation of the rails is l. The rails are connected to a battery so that a constant current I passes through the rod. The whole set-up is placed inside a uniform magnetic field B pointing vertically downwards. Assume the rod remains horizontal as it slides, and the rails are smooth.



What is the magnitude of the magnetic force F acting on the rod due to the magnetic field B? What is the direction of F as seen by the observer?

A BIl sin θ (up the rails)

B BIl sin θ (down the rails)

C BIl (towards left)

D BIl (towards right)

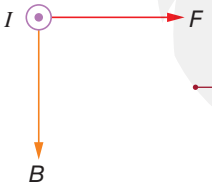
Solution

The rod is horizontal and perpendicular to B.

∴ Magnitude of $F = BIl$

Common mistake ⚠
Students may wrongly think that B makes an angle θ with I and wrongly apply the formula $F = BIl$ sin θ.

The current flows towards the observer. By Fleming's left-hand rule, F points to the right.



Common mistake ⚠
Students may not realize that the magnetic force must be perpendicular to the magnetic field and wrongly think that it is along the rails.

∴ The answer is D.

▶ Revision 6 Q11 (p.258)

Exercise 6.1

1 A current of 0.5 A flows through a straight wire. The strength of the Earth's magnetic field is 50 μT. Estimate the magnitude of the maximum magnetic force per unit length acting on the wire.

A 2.5×10^{-5} N B 10^{-4} N

C 25 N D 10^{4} N

★**2** A current-carrying straight wire L is placed at position P between two slab-shaped magnets (Fig a).

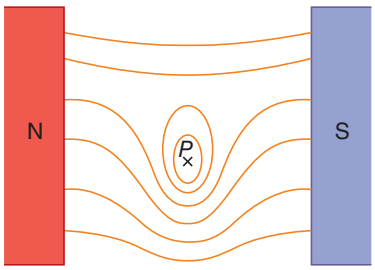


Fig a

Which of the following statements are correct?

(1) The current in L is pointing out of the page.

(2) The direction of magnetic force is pointing upwards.

(3) The magnetic field set up by L is anticlockwise.

A (1) and (2) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

★**3** A straight current-carrying conductor is in a uniform magnetic field B. Which of the following statements about the magnetic force acting on the conductor are correct?

(1) It must be perpendicular to magnetic field B.

(2) It must be perpendicular to the current.

(3) There is no magnetic force if the current is not perpendicular to magnetic field B.

A (1) and (2) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

★**4** A student sets up a current balance as shown (Fig b). When there is no current, the reading of the electronic balance is set to zero. Now, a current of 3 A flows from Y to X and the reading of the electronic balance becomes 0.87 g.

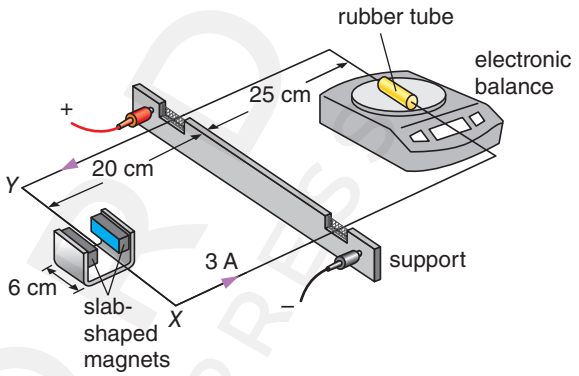


Fig b

(a) Find the magnetic force F_B acting on XY.

(b) Is the blue face of the magnet an N-pole or S-pole?

(c) Estimate the strength of the magnetic field in the region within the slab-shaped magnets.

★**5** Figure c shows a current-carrying conductor in a uniform magnetic field. The conductor is 10 cm long.

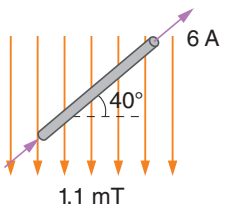


Fig c

(a) In the figure, mark the direction of the magnetic force acting on the conductor.

(b) Find the magnitude of the magnetic force acting on the conductor.

(c) How will the answer in (a) and (b) change if the conductor is rotated clockwise as shown in Figure d?

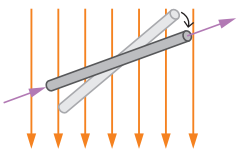


Fig d

6.2 D.c. motors

1 Turning effect on a coil

DSE goal

Determine turning effect on current-carrying coil in magnetic field.

Simulation

→ This simulation shows the forces acting on the two sides of a current-carrying coil and the resultant turning effect. Ss can change the current direction and magnet polarity.

In Figure 6.2a, wire WX carries a current I and is perpendicular to a uniform magnetic field B. It experiences an upward magnetic force of magnitude F. Wire ZY carries a current I in the opposite direction and experiences a downward magnetic force of magnitude F. If wires WX and ZY are free to move, they will move upwards and downwards respectively.

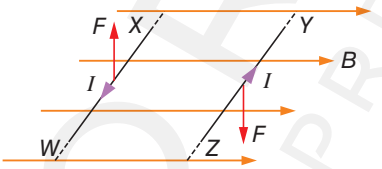


Fig 6.2a *Magnetic forces acting on current-carrying wires.*

What will happen if the two wires are connected to form a rectangular coil $WXYZ$? As shown in Figure 6.2b, an upward magnetic force and a downward magnetic force of the same magnitude act on WX and YZ respectively. These forces produce a turning effect on the coil.

What is the direction of magnetic force acted on XY and WZ?

If the coil is vertical, the magnetic forces acting on WX and YZ will have no turning effect (Fig 6.2c).

May ask Ss to draw the directions of the magnetic forces acting on XY and WZ when the coil is vertical.

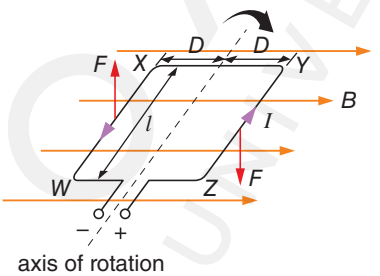


Fig 6.2b *Magnetic forces acting on a current-carrying coil produce a turning effect.*

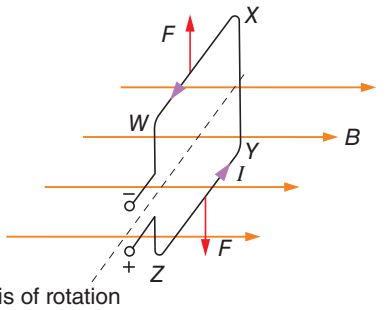


Fig 6.2c *No turning effect if the coil is vertical.*

If the coil has N turns and the length of side WX is l, there will be N conductors (wires) of length l carrying current I in the same direction at side WX. Therefore, the overall force acting on side WX is given by

$$F = BIl \times N = BIlN$$

The force acting on side YZ has the same magnitude.

Suppose WX and YZ are at the same distance D from the axis of rotation. When the coil is horizontal, the moment τ about the axis due to the forces acting on WX and YZ is

$$\tau = Fd_{\perp} \times 2 = (BIlN)(D) \times 2 = BI(2Dl)N = BIAN$$

where $A = 2Dl$ is the area of the coil.

This is the maximum turning effect on the coil.

Example 5 Calculating the turning effect on a coil

A coil of N turns and an area of A ($= l \times D$) is placed in a uniform magnetic field B (Fig a). The axis of rotation of the coil is perpendicular to B. The current flowing through the coil is I. Suppose the normal of the coil makes an angle ϕ to B (Fig b). The forces acting on WX and YZ has the same magnitude F.

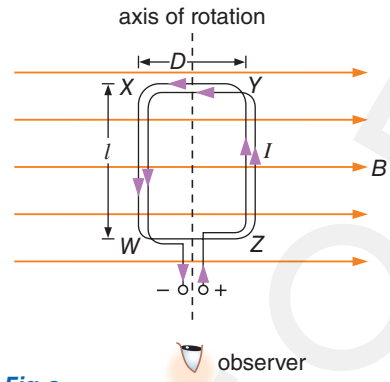


Fig a

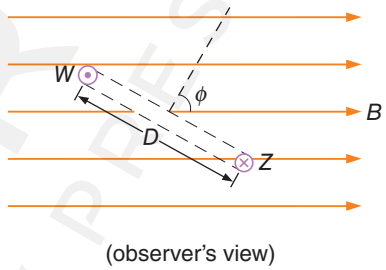


Fig b

(a) Draw in Figure b the directions of the magnetic forces acting on side WX and YZ.

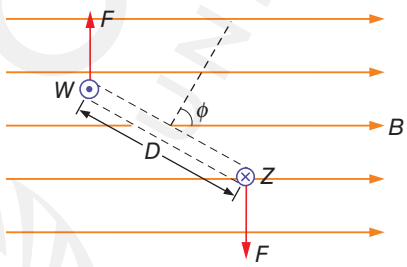
(b) Find F.

(c) Find the total moment acting on the coil.

(d) How does the total moment change when ϕ decreases?

Solution

(a)



(b) $F = BIl \times N = BIlN$

(c) Take moment about the axis of rotation. By $\tau = Fd_\perp$,

$$\text{total moment} = \left(F \times \frac{D}{2} \sin\phi\right) \times 2$$

$$= (BIlN) \times D \sin\phi$$

$$= BI(Dl)N \sin\phi$$

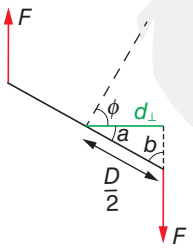
$$= BIAN \sin\phi \text{ (clockwise)}$$

(d) Decreases

Caution

The angle between B and I is 90°, not ϕ.

Tip



$a = 90° - \phi$

$b = 180° - 90° - a = \phi$

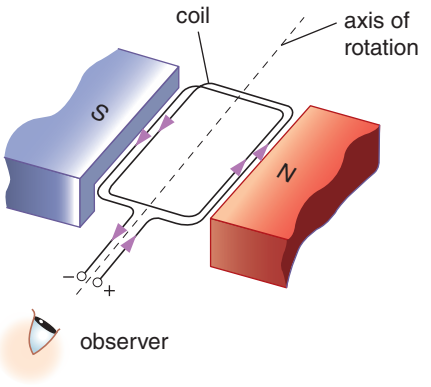
$d_\perp = \frac{D}{2} \sin b = \frac{D}{2} \sin\phi$

Google Forms Microsoft Forms

Quick check 4

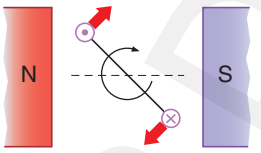
→ E-version in OUP Exercise Platform is also available.

1 In the figure below, as seen from the perspective of the observer, the moment acting on the coil is in (clockwise / anticlockwise) direction.

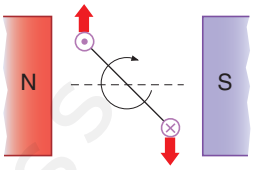


2 Which of the following diagrams correctly shows the directions of the magnetic forces (indicated by thick arrows) acting on the coil?

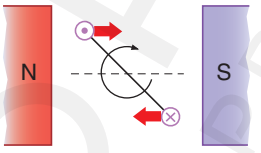
A



B



C



D None of the above

Learn by practice 2

1 A 15-turn coil of a size 12 cm by 16 cm is shown in Figure a. Find the moment acting on the coil about the axis of rotation

(a) when the plane of the coil is parallel to the magnetic field (Fig b);

(b) when the normal of the coil makes an angle of 60° to the magnetic field (Fig c).

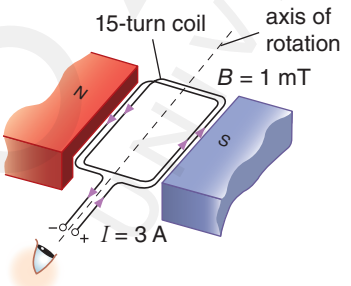


Fig a

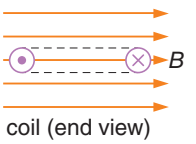


Fig b

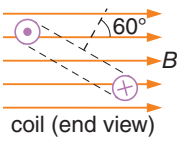


Fig c

DSE goal

Describe structure of simple d.c. motor and how it works.

DSE exam

17(1A)Q26, 19(1A)Q27

2 How a d.c. motor works

An **electric motor** is a device making use of the turning effect of a coil inside a magnetic field. It converts electrical energy into kinetic energy and is commonly found in many electrical appliances such as electric fans and washing machines.

An electric motor is basically a current-carrying coil rotating continuously in a magnetic field. However, a simple coil cannot keep rotating in a magnetic field in a practical situation. Let us find out why.

electric motor 電動機

Consider the set-up in Figure 6.2d. The coil is initially horizontal.

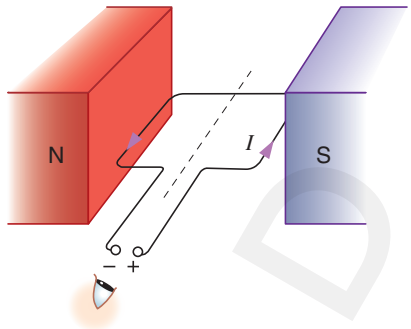


Fig 6.2d *A horizontal coil in a uniform magnetic field.*

When the power supply is switched on, a current I flows through the coil, so the coil starts rotating clockwise (Fig 6.2e ①, ②).

The direction of the force acting on each wire can be found by using Fleming's left-hand rule. ▶

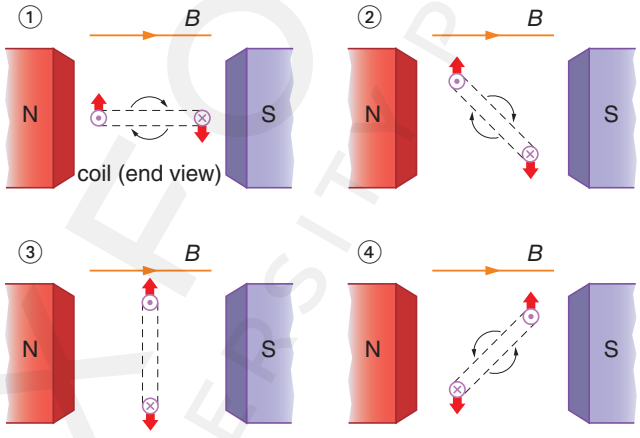


Fig 6.2e *The turning effect on a coil in a magnetic field.*

When the coil is vertical (③), the magnetic forces have no turning effect. The coil shoots past the vertical by inertia. After that, the turning effect reverses (④), the coil turns back, and shoots past the vertical again. As a result, the coil vibrates about the vertical position and cannot rotate continuously.

May encourage Ss to list out the problems without a commutator. Guide Ss to include the following key points:
1 The coil cannot turn 360°.
2 If it can, then twisting of wire occurs.

To allow the coil in a motor to rotate continuously, some modifications are needed. Figure 6.2f shows the structure of a simple electric motor.

Video

→ This video demonstrates how to make a simple motor.

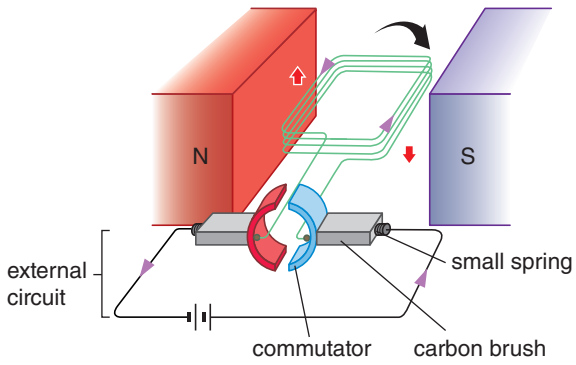


Fig 6.2f *A simple d.c. motor*

A commutator is also called a split ring.

In a simple d.c. motor, a **commutator**, which is a metal ring split into two halves, is fixed to the coil and rotates with it. A pair of **carbon brushes** are pushed against the commutator by two small springs. The carbon brushes are connected to the power supply.

Simulation

→ This simulation allows Ss to change the size of the current, the number of turns in the coil, the area of the coil and the strength of the magnets, to see how the turning effect of a motor is affected.

Figure 6.2g shows how a d.c motor works.

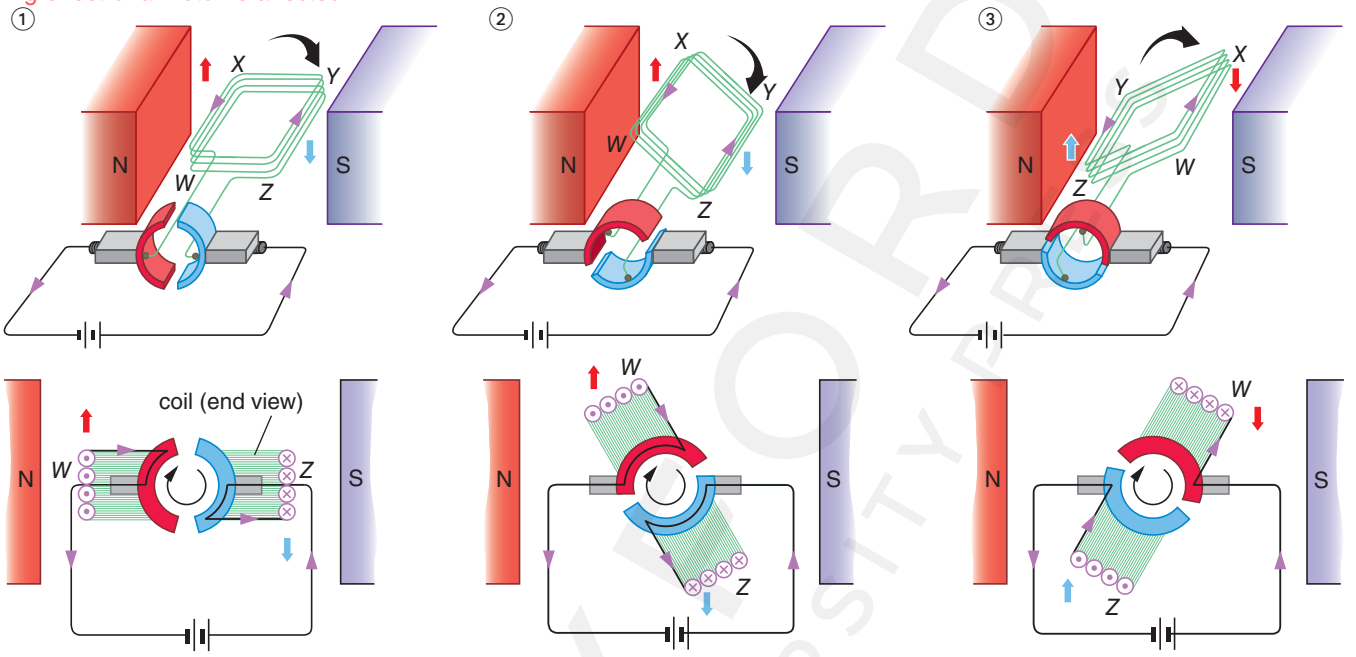


Fig 6.2g *The commutator changes the current direction in the coil as the coil shoots past the vertical.*

A shortcoming of the commutator is that sparks may be produced between the commutator and the carbon brushes during rotation. This causes the parts to wear out more easily.

① The coil turns when current flows into it.

② Every half turn when the coil shoots past the vertical position due to inertia, the commutator changes contact from one brush to the other.

③ The current in the coil and hence the magnetic forces reverse in direction. Therefore, the coil keeps rotating in the same direction.

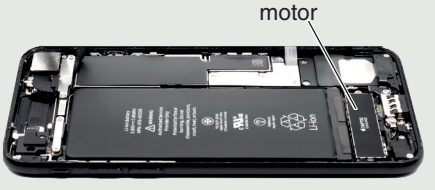
A d.c. flows in the external circuit and an a.c. flows in the coil.

In a simple d.c. motor, a direct current flows into the coil via a commutator. Magnetic forces act on the coil and make it rotate. The commutator reverses the current in the coil every half turn and keeps the coil rotating in the same direction.

Know more

Motors in smartphone

All of us carry a motor with us in in our smartphone. Do you know what the function of this motor is? (Hint: When is your mobile phone 'in motion'?)



The motor is used in 'vibrating mode'.

commutator 換向器　　carbon brush 碳電刷

Know more

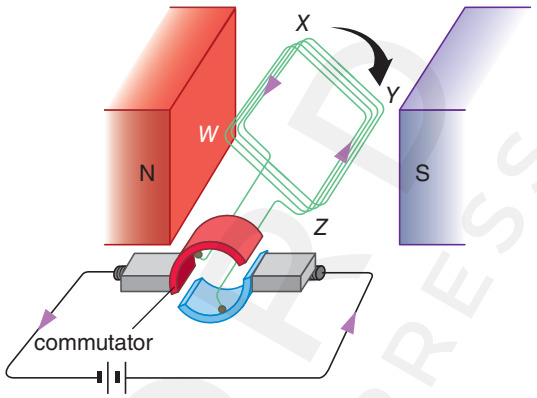
Practical motors

Practical motors have some features to strengthen the turning effect.

▶ p.266

Example 6 A simple d.c. motor

In a simple motor, the coil rotates in a uniform magnetic field.



(a) When the coil rotates,

(i) how does the force acting on side *YZ* change;

(ii) how does the moment acting on the coil change?

(b) State three methods to increase the turning speed of this motor.

Solution

(a) (i) The direction of the force reverses every time the coil shoots pass the vertical position.

The magnitude of the force remains unchanged.

(ii) The direction of the moment remains unchanged.

The magnitude of the moment is the maximum when the coil is horizontal. It decreases when the coil rotates and becomes zero when the coil is vertical.

(b) Increase the number of turns of the coil. / Increase the area of the coil. / Increase the current flowing through the coil. / Use a stronger magnet.

May tell Ss that the turning effect can also be increased by using curved magnets instead of flat ones or adding a soft-iron core to the coil.

Tip

By $F = BIl \sin\theta$, since B, I and l are all constant and θ is always 90°, the magnitude of F is constant.

Tip

From Example 5 (p.235), $\tau = BIAN \sin\phi$

When the coil is horizonal, $\phi = 90°$ and τ is max.

When the coil is vertical, $\phi = 0°$ and $\tau = 0$.

Tip

By $\tau = BIAN \sin\phi$, the turning effect of a motor increases when B, I, A or N increases.

Quick check 5

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 In a d.c. motor, a (carbon brush / commutator) is fixed to the coil. It reverses the current flowing in the coil every half cycle. It is connected to the external circuit via a pair of (carbon brushes / commutators).

2 Which of the following can increase the turning effect of a d.c. motor? Put a tick in the corresponding box(es).

- ☐ Increasing the current
- ☐ Increasing the number of turns of the coil
- ☐ Using a stronger magnet
- ☐ Increasing the area of the coil

☆ HKDSE 2019 Paper 1A Q27

Example 7 A simple motor with electromagnets

A student sets up a motor as shown (Fig a). *AB* and *CD* are solenoids connected to an a.c. supply.

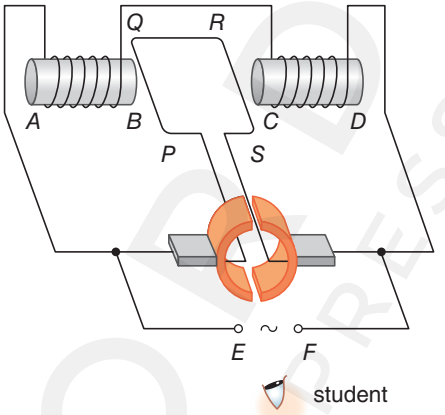


Fig a

Figure b shows how the output voltage *V* of the a.c. supply varies with time *t*. Positive voltage indicates that the potential at *E* is higher than that at *F*.

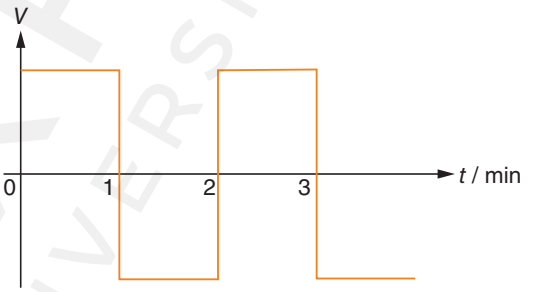


Fig b

(a) From $t = 0$ to $t = 1$ min,

(i) what are the polarities of ends *B* and *C* of the solenoids?

(ii) how does the coil *PQRS* move as seen by the student?

(b) From $t = 1$ min to $t = 2$ min, are the polarities of ends *B* and *C*, and the motion of the coil the same as that in (a)?

(c) Does the coil rotate continuously if *V* varies at 50 Hz?

Solution

(a) (i) *B* is an S-pole while *C* is an N-pole.

(ii) It rotates clockwise.

(b) *B* becomes an N-pole while *C* becomes an S-pole.

The coil keeps rotating clockwise.

(c) Yes

Tip

The current in the coil also reverses, so the direction of rotation of the coil remains the same.

Revision 6 Q10 (p.258)

Exercise 6.2

1 An incomplete structure of a d.c. motor is shown below (Fig a).

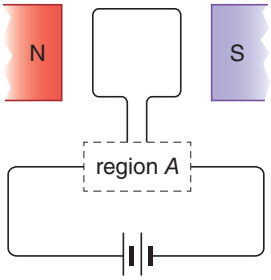


Fig a

What should be used in region *A* to keep the coil rotating in one direction?

(1) A pair of carbon brushes

(2) A soft-iron core

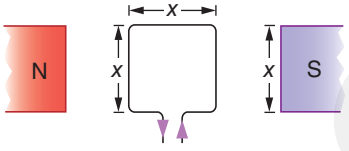
(3) A commutator

A (1) only B (1) and (3) only

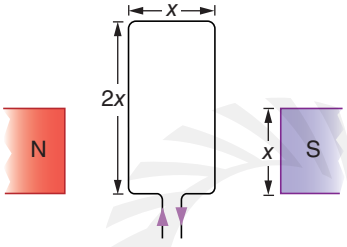
C (2) and (3) only D (1), (2) and (3)

★**2** In each of the following cases, a rectangular coil is placed between two magnets. In which case does the coil have the largest turning effect?

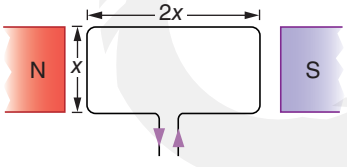
A



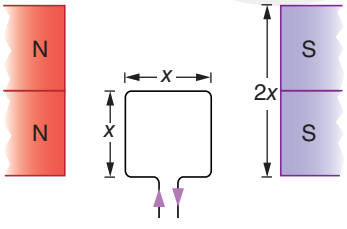
B



C



D



★**3** Which of the following about a d.c. motor is/are correct?

(1) The carbon brushes rotate with the coil.

(2) The commutator reverses the connections of the coil with the external circuit every half turn.

(3) The current in the coil changes direction periodically.

A (1) only B (2) only

C (2) and (3) only D (1), (2) and (3)

★**4** Figure b shows the design of a d.c. motor. The coil has 3 turns and a dimension of 30 cm × 10 cm. *A* and *B* are two unlike magnetic poles. The magnetic flux density between *A* and *B* is 0.05 T. The current passing through the coil is 3 A. The motor rotates anticlockwise when observed at point *P*.

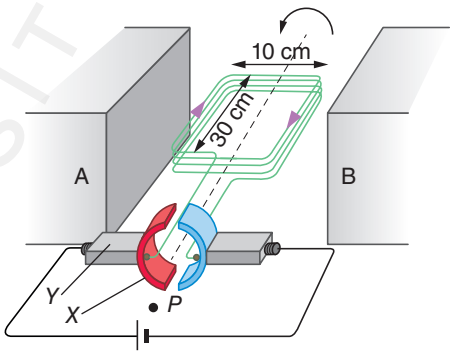


Fig b

(a) Which pole, *A* or *B*, is the N-pole?

(b) Find the maximum moment acting on the coil.

(c) Name parts *X* and *Y* and state their functions.

★**5** A rectangular coil of 20 turns and an area of 30 cm^2 is placed in a uniform magnetic field *B* of 2 mT. The current flowing through the coil is 5 A. Figure c shows the end view of the coil.

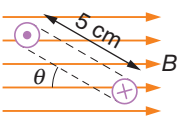


Fig c

(a) In Figure c, mark the directions of the forces acting on the two sides of the coil.

(b) (i) What is the direction of the moment?

(ii) Find the moment τ acting on the coil when $\theta = 30°$.

(c) Suggest a method to increase the turning effect on the coil.

6.3 More about magnetic forces

DSE goal

Quantitative treatment of force between currents in long straight parallel conductors.

DSE exam

12(1A)Q29, 17(1B)Q9, 19(1A)Q26, 22(1A)Q27

1 Two long straight parallel wires

In Figure 6.3a, two long straight parallel wires P and Q are r apart. They carry currents I_P and I_Q in opposite directions. I_P sets up a magnetic field B_P which causes Q to experience a magnetic force F_{QP} pointing away from P (Fig 6.3b). Similarly, P experiences a magnetic force F_{PQ} pointing away from Q (Fig 6.3c). As a result, P and Q repel each other.

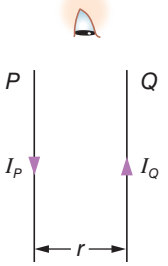


Fig 6.3a *P and Q carrying currents in opposite directions.*

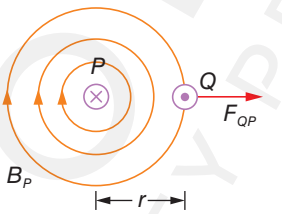


Fig 6.3b *The magnetic force acting on Q due to P.*

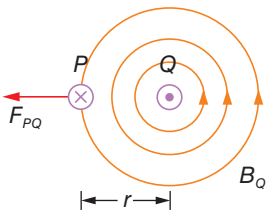


Fig 6.3c *The magnetic force acting on P due to Q.*

Conversely, if I_P and I_Q are in the same direction, P and Q will attract each other (Fig 6.3d–6.3f).

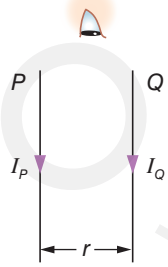


Fig 6.3d *P and Q carrying currents in the same direction.*

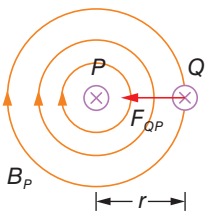


Fig 6.3e *The magnetic force acting on Q due to P.*

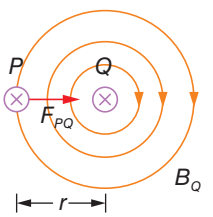


Fig 6.3f *The magnetic force acting on P due to Q.*

Two wires repel each other if they carry currents in opposite directions. They attract each other if they carry currents in the same direction (Fig 6.3g).

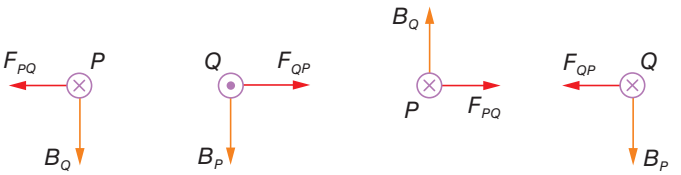


Fig 6.3g *Forces between two wires.*

Let us find out the magnitude of the force acting on each wire. At Q,

Cross-link
You learned the magnetic field produce by a long straight wire in Ch 5.2.

$$B_P = \frac{\mu_0 I_P}{2\pi r}$$

Consider a section of Q with length L. By $F = BIl$,

$$F_{QP} = B_P I_Q L = \frac{\mu_0 I_P I_Q L}{2\pi r}$$

$$\therefore \text{ Force per unit length acting on } Q = \frac{F_{QP}}{L} = \frac{\mu_0 I_P I_Q}{2\pi r}$$

Try to derive the force per unit length acting on P by considering the magnetic field B_Q produced by I_Q.

Since the force acting on Q by P and the force acting on P by Q are an **action-and-reaction pair**, the force per unit length acting on P is also $\frac{\mu_0 I_P I_Q}{2\pi r}$ but in the opposite direction.

For two long straight parallel wires *P* and *Q*,

permeability of free space = $4\pi \times 10^{-7}$ T m A^{-1}

current in wire *P* (A)

current in wire Q (A)

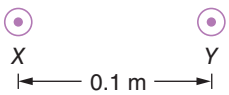
$$\textbf{force per unit length} = \frac{\mu_0 I_P I_Q}{2\pi r}$$

unit: N m^{-1}

distance between wires (m)

Example 8 Interaction between two parallel wires

Two long straight parallel wires X and Y carry currents out of the page. They are 0.1 m apart. The currents in X and Y are 3 A and 2 A respectively. Take $\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1}.



(a) Find the magnetic force per unit length acting on X by Y.

(b) Find the magnetic force per unit length acting on Y by X.

Solution

Pay attention to the unit.

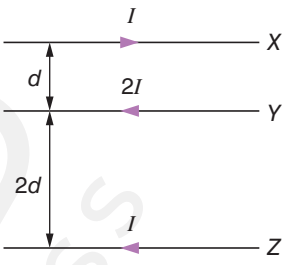
(a) Force per unit length $= \frac{\mu_0 I_X I_Y}{2\pi r} = \frac{(4\pi \times 10^{-7})(3)(2)}{2\pi(0.1)} = 1.2 \times 10^{-5}$ N m^{-1}

Since the currents are in the same direction, the wires attract each other and the force acting on X points to the right.

(b) By Newton's third law, the force per unit length acting on Y is 1.2×10^{-5} N m^{-1} (to the left).

Example 9 | Interaction between three parallel wires

Three long straight parallel wires, X, Y and Z, carry currents of I, $2I$ and I in the directions as shown. The distance between X and Y is d and the distance between Y and Z is $2d$. The magnitude of the magnetic force per unit length acting on Y due to X is F. Find the resultant magnetic force per unit length $\frac{F_Y}{L}$ acting on Y due to both X and Z.



Solution

$$\frac{F_{YX}}{L} = \frac{\mu_0 I_X I_Y}{2\pi r} = \frac{\mu_0 (I)(2I)}{2\pi (d)} = \frac{\mu_0 I^2}{\pi d} = F \text{ (downwards)}$$

$$\frac{F_{YZ}}{L} = \frac{\mu_0 I_Z I_Y}{2\pi r} = \frac{\mu_0 (I)(2I)}{2\pi (2d)} = \frac{1}{2} \times \frac{\mu_0 I^2}{\pi d} = \frac{F}{2} \text{ (downwards)}$$

$$\frac{F_Y}{L} = F + \frac{F}{2} = \frac{3F}{2} \text{ (downwards)}$$

Quick check 6

→ E-version in OUP Exercise Platform is also available.

Google Forms | Microsoft Forms

Take $\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1}.

1 Two long straight parallel wires X and Y are separated by r. Complete the following table about the magnitude of the magnetic force per unit length F between the wires.

	I_X / A	I_Y / A	r / m	F / N m^{-1}
(a)	0.1	2		8×10^{-7}
(b)	3		0.02	6×10^{-6}

2 Currents of 2 A and 5 A flow along two long straight parallel wires X and Y in opposite directions (Fig a). Determine whether the following statements are true.

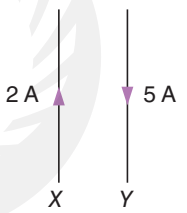


Fig a

(a) The magnetic force acting on X by Y is larger than that acting on Y by X. (T/F)

(b) The magnetic force acting on X by Y points to the left. (T/F)

3 Three long straight parallel wires X, Y and Z are arranged as shown (Fig b). Each wire carries a current I flowing in the directions indicated.

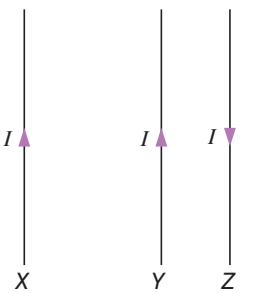


Fig b

The resultant magnetic force per unit length acting on Y is F. Determine whether the following statements are true.

(a) F points to the left. (T/F)

(b) F is smaller than the magnetic force per unit length acting on Y by X. (T/F)

Learn by practice 3

Take $\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1}.

1 Currents flow along two long straight parallel wires. In each case,

(a) draw the directions of the magnetic field B and the force F acting on each wire;

(b) compare the forces that the wires act on each other.

(i) (ii)



2 Two parallel long conductors X and Y are connected to an a.c. power supply (Fig a). Which of the following statements is correct?

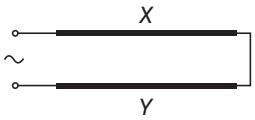


Fig a

A X and Y always attract each other.

B X and Y always repel each other.

C X and Y attract and repel each other alternatively.

3 Figure b shows three current-carrying wires P, Q and R.

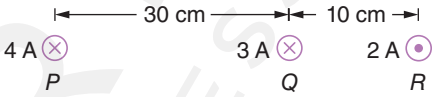


Fig b

Find the magnetic force per unit length acting on P by

(a) Q;

(b) R;

(c) both Q and R.

Ext

DSE goal

Represent force on moving charge in magnetic field by $F = BQv \sin\theta$ and solve problems.

DSE exam

13(1B)Q4, 18(1A)Q28, 23(1A)Q30

2 Moving charges in a magnetic field

a Magnetic forces on moving charges

We have learned that a magnetic force acts on a current-carrying conductor in a magnetic field. In fact, this magnetic force is the sum of the force acting on each moving charge in the conductor.

Consider a charge Q moving at velocity v in a direction **perpendicular** to a magnetic field B. The magnitude of the magnetic force F experienced by the charge is given by

$$F = BQv$$

The direction of F can be found by using Fleming's left-hand rule. The direction of current is taken as the moving direction of a positive charge, and opposite to the moving direction of a negative charge (Fig 6.3h).

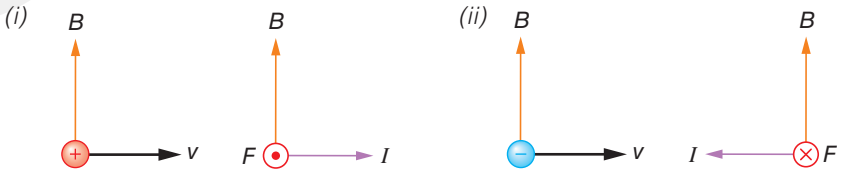


Fig 6.3h *Applying Fleming's left-hand rule on (i) a positive charge and (ii) a negative charge.*

Know more

Deriving $F = BQv$ from $F = BIl$

Do you know how can the equation $F = BQv$ on a moving charge be derived from the equation $F = BIl$ on a conductor?

▸ p.265

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Ext

When the charge moves in a direction making an angle θ with the magnetic field B, only the perpendicular component of B (i.e. $B \sin \theta$) contributes to the magnetic force F on the charge particle (Fig 6.3i).

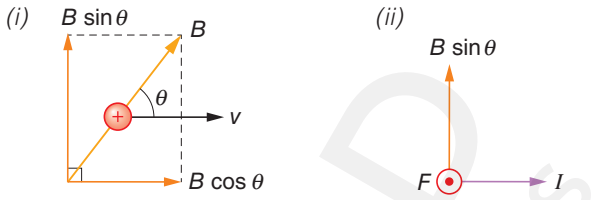
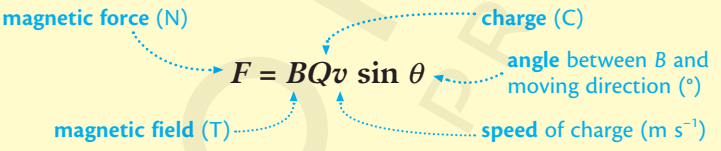


Fig 6.3i *(i) A positive charge moving at an angle θ to a magnetic field B and (ii) a magnetic force acts on the charge.*

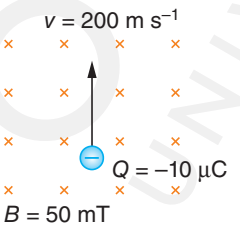


If the charge moves parallel to the magnetic field, θ will be 0° and F will be zero.

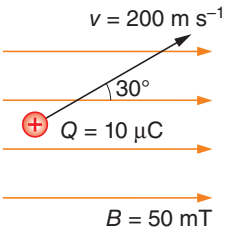
Example 10 | A charge moving in a magnetic field

Find the magnitude and direction of the magnetic force F acting on the moving charge in the following cases.

(a)



(b)



Solution

(a) $F = BQv$
$= (50 \times 10^{-3})(10 \times 10^{-6})(200)$
$= 10^{-4}$ N

Since the direction of I is opposite to the moving direction of a negative charge, by Fleming's left-hand rule, F points to the right.

(b) $F = BQv \sin \theta$
$= (50 \times 10^{-3})(10 \times 10^{-6})(200) \sin 30°$
$= 5 \times 10^{-5}$ N

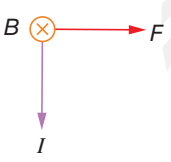
By Fleming's left-hand rule, F points into the page.

Tip

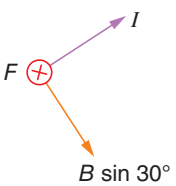
In calculating the magnitude of F, ignore the sign of Q.

The direction of F:

(a)



(b)



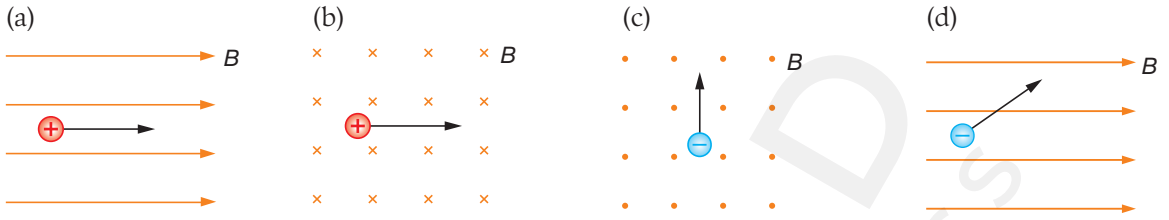
Ext

Quick check 7

→ E-version in OUP Exercise Platform is also available.

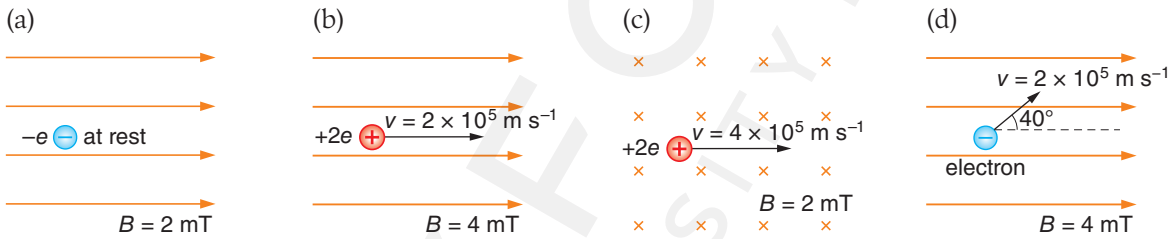
Google Forms Microsoft Forms

1 Charged particles move in different magnetic fields as shown. State the direction of the magnetic force acting on each particle.



Learn by practice 4

1 In each case, find the magnitude of the magnetic force acting on the charge and draw the direction of the force in the figure. Take $e = 1.60 \times 10^{-19}$ C.



b Motion of a charge in a magnetic field

We do not consider the effect of gravity unless otherwise specified.

May tell more able Ss that if v makes an angle with B, the charge will move along a helix path.

Consider a charge Q of mass m moving with an initial speed v perpendicular to a uniform magnetic field B (Fig 6.3j). It experiences a magnetic force F perpendicular to its moving direction. This changes its moving direction. Since F is always perpendicular to the moving direction, it does no work on the charge. The charge keeps moving with the same speed and the magnitude of F remains unchanged. As a result, the charge performs uniform circular motion.

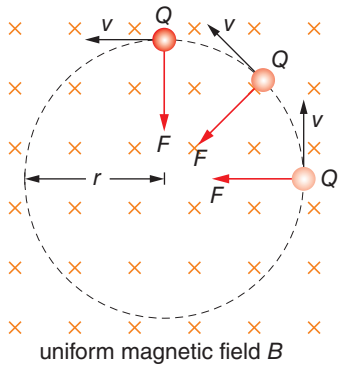


Fig 6.3j *A charged particle undergoes circular motion in a uniform magnetic field.*

As the magnetic force F provides the centripetal force, we have

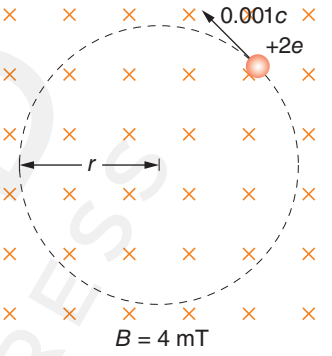
$$BQv = \frac{mv^2}{r}, \text{ where } r \text{ is the radius of the circular path.}$$

If a charge moves parallel to a magnetic field, no magnetic force will act on it, so it will continue to move along a straight line.

Ext

Example 11 Circular motion of a charge particle in a magnetic field

A particle moves in a circular path in a uniform magnetic field at a speed of $0.001c$. It has a mass of 6.64×10^{-27} kg and carries a charge of $+2e$. The strength of the magnetic field B is 4 mT. Take $e = 1.60 \times 10^{-19}$ C and $c = 3.00 \times 10^{8}$ m s^{-1}. Calculate the radius r of the circular path.



Solution

As the magnetic force provides the centripetal force,

By $BQv = \dfrac{mv^2}{r}$,

$$r = \frac{mv}{BQ} = \frac{(6.64 \times 10^{-27})(0.001 \times 3.00 \times 10^{8})}{(4 \times 10^{-3})(2 \times 1.60 \times 10^{-19})} = 1.56 \text{ m}$$

Quick check 8

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 Which of the following can be the effect of a magnetic field on a moving charge? Put a tick in the corresponding box(es)

- ☐ Changing the moving direction of a charge
- ☐ Slowing down a moving charge
- ☐ Speeding up a moving charge
- ☐ Having no effect on a moving charge

2 A charged particle Q is moving in a uniform magnetic field. Is it moving (A) parallel, or (B) perpendicular to the field?

(a) Q accelerates.

(b) Q moves at a constant velocity.

(c) Q moves in a straight path.

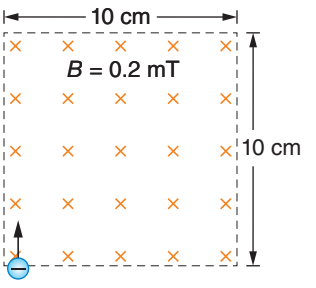
(d) Q moves in a circular path.

Learn by practice 5

1 An electron enters a uniform magnetic field of 0.2 mT perpendicularly at 1.75×10^{6} m s^{-1}. The electron has a mass of 9.11×10^{-31} kg and carries a charge of -1.60×10^{-19} C.

(a) The electron will move in a circular arc (part of a circle) inside the magnetic field. Find the radius of the circular arc.

(b) Sketch the path of the electron inside the magnetic field in the figure.



Ext

☆ HKDSE 2013 Paper 1B Q4

Example 12 Two charged particles moving in a magnetic field

Two particles X and Y enter a magnetic field perpendicularly and move along circular paths. Their details are given below.

	Speed	Mass	Charge
X	v	m	Q
Y	$2v$	m	Q

(a) Find the ratio of
(i) the radius of the path of X to that of Y,
(ii) the period of X to that of Y.

(b) Explain how the speed of X changes when it moves in the magnetic field.

Solution

(a) (i) By $BQv = \dfrac{mv^2}{r}$,

$$r = \frac{mv}{QB} \propto v \quad (m, Q \text{ and } B \text{ constant})$$

$$\therefore \frac{r_X}{r_Y} = \frac{v_X}{v_Y}$$

$$= \frac{v}{2v}$$

$$= \frac{1}{2}$$

(ii) $T = \dfrac{2\pi r}{v} \propto \dfrac{r}{v}$

$$\therefore \frac{T_X}{T_Y} = \frac{r_X}{r_Y} \times \frac{v_Y}{v_X}$$

$$= \frac{1}{2} \times \frac{2v}{v}$$

$$= 1$$

(b) The magnetic force acting on X is always perpendicular to the moving direction of X and hence does no work on X. Therefore, the speed of X remains unchanged.

▶ Exercise 6.3 Q6 (p.253)

Tip
The faster the particle, the larger the radius of its path.

Tip
The periods are the same even if the particles have different speeds.
Alternatively,
$T = \dfrac{2\pi r}{v} = \dfrac{2\pi}{v} \times \dfrac{mv}{BQ} = \dfrac{2\pi m}{BQ}$
$\therefore T$ is independent of v.

Think about...

If a particle Z of speed v, mass $2m$ and charge Q enters the magnetic field perpendicularly, what will be the ratio of its period to that of X?

Ext

DSE exam

12(1A)Q30, 20(1A)Q27, 22(1A)Q29

C Motion of a charge in both an electric field and a magnetic field

Sometimes, a charged particle moves in the presence of both a magnetic field and an electric field. In Figure 6.3k, a positive charge Q of mass m moves with a speed v inside a uniform electric field of strength E and a uniform magnetic field of strength B. The electric field points downwards and the magnetic field points into the page.

If the particle is negatively charged, F_B will point downwards and F_E will point upwards. ▶

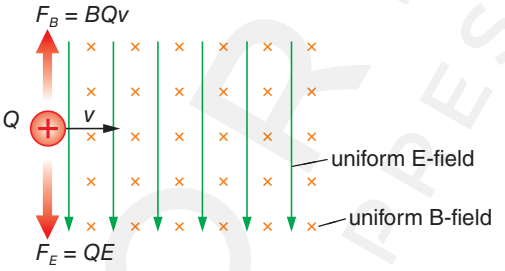


Fig 6.3k *A charged particle moves in a region with a magnetic field and an electric field perpendicular to each other.*

Know more

Charge-to-mass ratio

In 1897, when electrons were still new to scientists, *J. J. Thomson* determined the charge-to-mass ratio (e/m) of an electron by applying both a magnetic field and an electric field to a cathode ray. This discovery helped scientists find the mass of an electron which is difficult to measure directly.

The magnetic force F_B acting on the particle is BQv and the electric force F_E is QE. F_B and F_E act in opposite directions.

1 If $F_B = F_E$, the particle will travel along a straight line through the region (Fig 6.3l). This happens when

$$BQv = QE$$

$$v = \frac{E}{B}$$

By adjusting E and B, we can select charged particles with a particular velocity to pass the region without deflection. This velocity selector is used in electron microscopes and mass spectrometer.

In other words, a charged particle entering the region with speed $v = \frac{E}{B}$ will not be deflected and will travel with constant v. This speed is independent of the type and amount of charge that the particle carries.

v is independent of Q. ▶

When $v \uparrow$, $F_B \uparrow$ but F_E remains unchanged. ▶

The net force acting on the particle is not constant as the direction of the magnetic force keeps changing with the direction of motion of the particle. The motion is not a projectile motion nor a circular motion.

2 If a positively charged particle enters the region with another speed, it will be deflected upwards or downwards (Fig 6.3l).

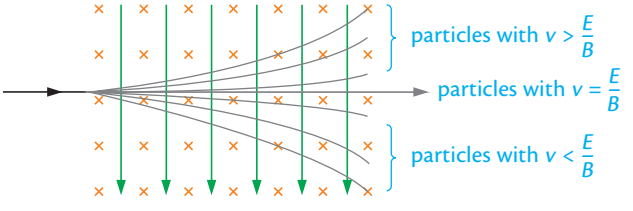


Fig 6.3l *A stream of positively charged particles entering a region of a magnetic field and an electric field at different speeds.*

Particles carrying opposite charges will be deflected in the opposite directions. ▶

Pay attention to the directions of the fields. If the magnetic field points into the page and the electric field points upwards, F_B and F_E will point in the same direction and will never cancel each other.

J. J. Thomson 湯姆生

Ext

Example 13 An electron in a magnetic field and an electric field

An electron flies horizontally into a uniform magnetic field of 5×10^{-3} T between two parallel metal plates at 2×10^{7} m s^{-1} as shown. The electron carries a charge of -1.60×10^{-19} C. The metal plates are separated by 0.01 m. A voltage is applied across the plates to make the electron travel horizontally between the plates.



(a) Find the direction of the electric field between the plates.

(b) Find the voltage across the two metal plates.

(c) If another electron flies horizontally into the field at a higher speed, how will it be deflected between the metal plates?

Solution

(a) By Fleming's left-hand rule, the magnetic force acting on the electron points downwards.

The electric force should point upwards to balance the magnetic force. Therefore, the electric field points downwards.

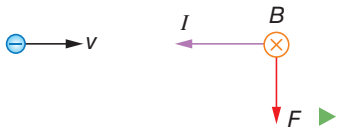
(b) The magnetic force is balanced by the electric force.

$$BQv = QE = Q\frac{V}{d}$$

$$V = Bvd = (5 \times 10^{-3})(2 \times 10^{7})(0.01) = 1000 \text{ V}$$

(c) An electron at a higher speed experiences a larger downward magnetic force but the same upward electric force. Therefore, the electron will be deflected downwards.

Tip



Tip

The electric field and the electric force acting on a negative charge are opposite in direction.

Cross-link

You learned $E = \frac{V}{d}$ in Ch 1.3.

Quick check 9

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 A positive charge moves along a straight line in a uniform magnetic field B between two parallel charged plates (Fig a). What is the direction of the electric field?

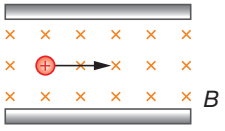


Fig a

2 A uniform magnetic field exists between two parallel charged plates. An electron moves in a straight line when passing through the region. What is the direction of the magnetic field?

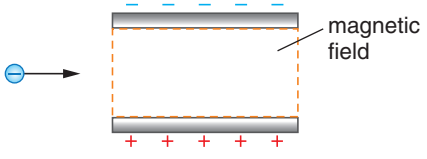
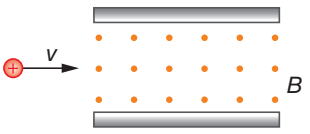


Fig b

Ext

Learn by practice 6

1 A particle carrying a charge of $+Q$ flies with velocity v into a uniform magnetic field B between two parallel charged plates. The electric field between the plates is E. The particle keeps moving along a straight line.



(a) State a relationship between B, E and v.

(b) How will the path of the particle change if

(i) it moves faster;

(ii) it has a greater mass;

(iii) it carries a charge of $-2Q$;

(iv) the separation between the plates decreases?

Exercise 6.3

Unless otherwise specified, assume that the effect of gravity is negligible.

★**1** X and Y are two current-carrying straight wires (Fig a). Wire Y is horizontal while wire X is vertical. O is a point on X and is on the same horizontal plane as Y.

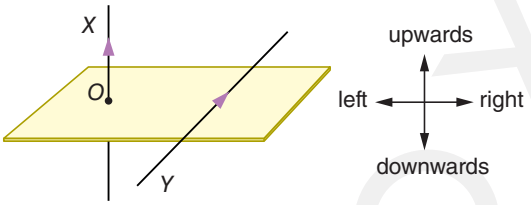


Fig a

What is the direction of the magnetic force acting on O by Y?

A Upwards

B Towards left

C Towards right

D No magnetic force

★**2** Ext A charge moving in a magnetic field experiences a magnetic force. Which of the following statements is **incorrect**?

A The direction of motion of the charge must be perpendicular to the field.

B The acceleration of the charge must be perpendicular to its direction of motion.

C The velocity of the charge is not constant.

D The speed of the charge is constant.

★**3** Long straight parallel wires P, Q, and R are arranged at the vertices of an equilateral triangle (Fig b). The currents flowing in them are the same in size and direction.

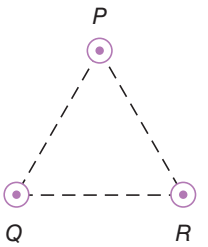


Fig b

What is the direction of the net magnetic force acting on wire P by the other two wires?

A Upwards B Downwards

C To the left D To the right

★**4** Ext Two charged particles X and Y enter a region between two parallel charged plates (Fig c). There is a uniform magnetic field pointing into the page in this region. The dotted lines in the figure show the paths of travel of the particles.

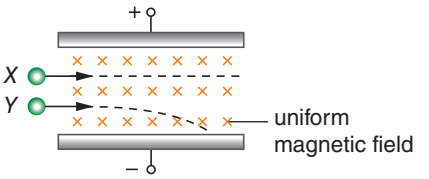


Fig c

Which of the following statements about the particles must be correct?

A Both of them carry positive charges.

B Y carries a larger amount of charges than X.

C X and Y enter the region at different speeds.

D X and Y have different masses.

★5 In Figure d, long straight parallel wires P, Q, and R lie on the same plane. Wire P carries a current of 3 A into the page while wire R carries the same amount of current out of the page. The resultant magnetic force per unit length acting on P due to Q and R is 1.8×10^{-6} N m^{-1} to the right. Find the size and direction of the current in Q.

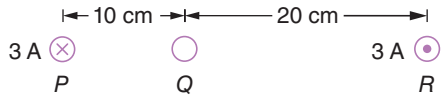


Fig d

★6 Ext A particle of charge $+e$ undergoes uniform circular motion inside a uniform magnetic field B as shown (Fig e).

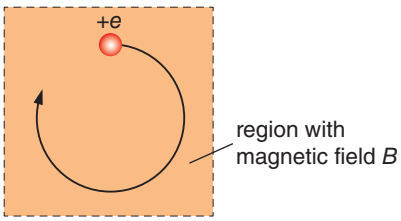


Fig e

(a) Determine the direction of B.

(b) How will the radius of the circular path change if the particle

(i) moves at a higher speed;

(ii) carries charge of $+2e$;

(iii) has a larger mass?

★7 Ext An electron moving to the right at 8.0×10^{6} m s^{-1} enters a uniform magnetic field of 2 mT (Fig f). The charge and the mass of an electron are -1.60×10^{-19} C and 9.11×10^{-31} kg respectively.

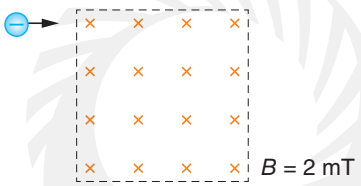


Fig f

(a) Find the radius of the path of the electron.

(b) Suppose the electron is moving to the left when it leaves the region.

(i) Sketch the path of the electron in Figure f.

(ii) How long will the electron stay in the region?

★8 Ext A charged particle travels through a region between two parallel charged plates without deflection (Fig g). There is also a uniform magnetic field B in this region.

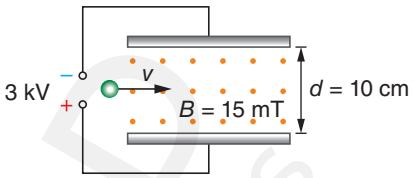


Fig g

(a) (i) Identify the directions of electric force and magnetic force if the particle is positively charged.

(ii) Does the particle carry positive or negative charge? Explain your answer.

(b) Find the speed of the particle.

(c) Suggest a way to deflect the particle.

★9 Ext A particle of charge Q and mass m performs uniform circular motion with speed v in a uniform magnetic field B (Fig h).

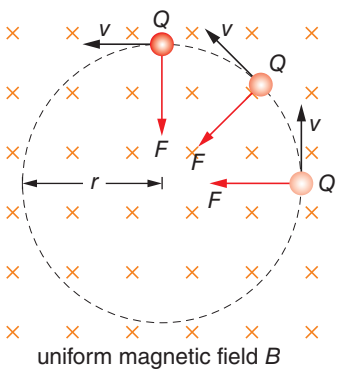


Fig h

(a) What is the magnitude of the magnetic force F acting on the charge?

(b) What is the radius r of the particle's path?

(c) Show that the time T for the particle to complete the circular path once is given by $T = \dfrac{2\pi m}{BQ}$.

(d) If both the speed v of the charged particle and the strength of the magnetic field B are doubled, how will F, r and T change?

Key concepts 6

1 Magnetic force on current-carrying conductor

- A current-carrying conductor experiences a magnetic force in a magnetic field.
- To find the direction of the magnetic force, Fleming's left-hand rule can be used (Fig a).
- To find the magnitude of magnetic force (Fig b), use:

  $F = BIl \sin \theta$

  where θ is the angle between the directions of B and I.

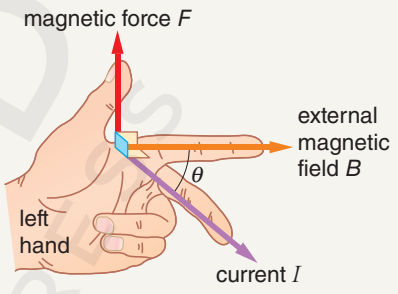


Fig a

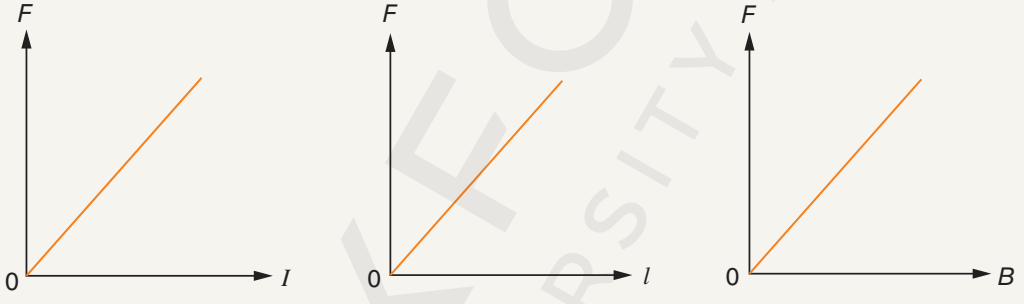


Fig b

2 Electric motors

- A current-carry coil in a magnetic field experiences a turning effect due to the magnetic forces acting on its two sides (Fig c).
- The commutator reverses the direction of the current in the coil to keep the coil rotating continuously in the same direction (Fig d).

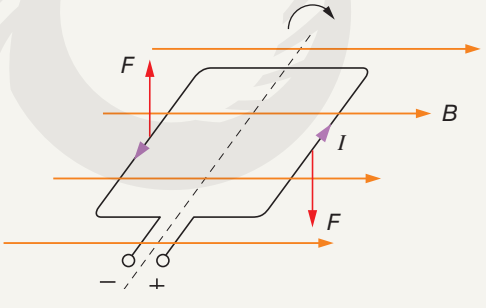


Fig c

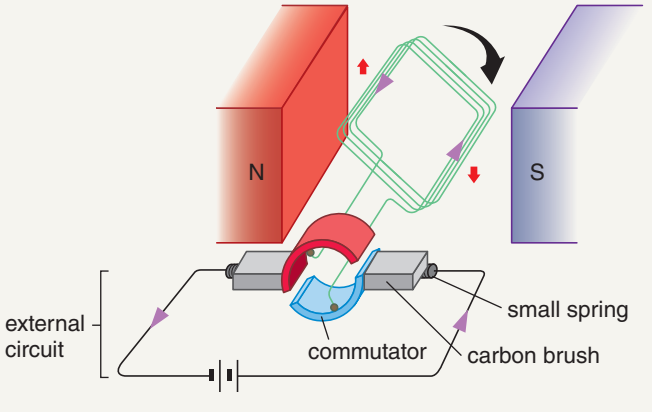


Fig d

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3 Magnetic forces between two parallel wires

- Two wires repel each other if they carry currents in opposite directions (Fig e). They attract each other if they carry currents in the same direction (Fig f).
- Force per unit length = $\frac{\mu_0 I_P I_Q}{2\pi r}$

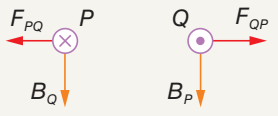


Fig e

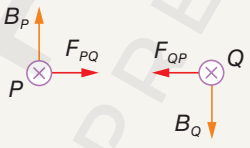


Fig f

4 Moving charge in magnetic field Ext

- To find the direction of magnetic force, Fleming's left-hand rule can be used but note that the direction of I is
  - the same as the moving direction of a positive charge
  - opposite to the moving direction of a negative charge
- To find the magnitude of magnetic force (Fig g), use:
  $F = BQv \sin\theta$
- If the initial direction of motion is perpendicular to B,
  - the magnetic force will do no work
  - the charge will move in uniform circular motion (Fig h)

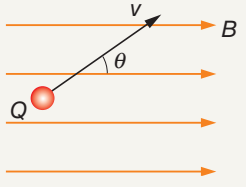


Fig g

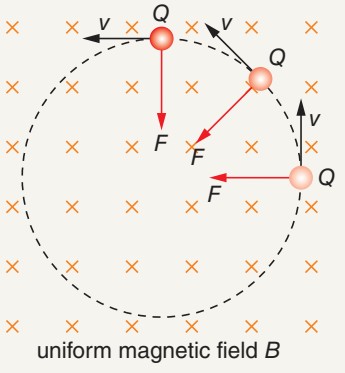


Fig h

Key terms

1	carbon brush 碳電刷	p.238	3	electric motor 電動機	p.236
2	commutator 換向器	p.238	4	Fleming's left-hand rule 弗林明左手定則	p.223

Revision 6

Unless otherwise specified, assume that the effect of gravity is negligible.

Multiple-choice questions

★1 6.2 The structure of a d.c. motor is shown in Figure a. The motor consists of two parts: the electromagnet and the rotator (coil *Y*).

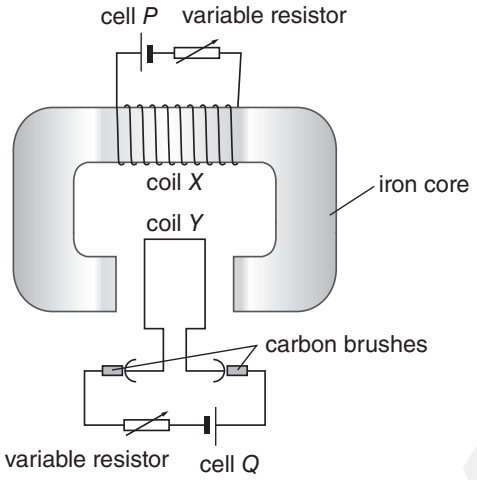


Fig a

Which of the following actions can reverse the direction of rotation of coil *Y*?

(1) Reverse the polarity of cell *P*.

(2) Reverse the polarity of cell *Q*.

(3) Reverse the polarity of both cells *P* and *Q*.

A (1) only
B (1) and (2) only
C (2) and (3) only
D (1), (2) and (3)

★2 Ext 6.3 An electron enters a uniform magnetic field. Its direction of motion makes an angle larger than 0° with the magnetic field. Which of the following statements about the electron is/are correct?

(1) The kinetic energy of the electron remains unchanged.

(2) It continues to travel with acceleration.

(3) Its direction of motion will eventually become perpendicular to the magnetic field.

A (2) only
B (3) only
C (1) and (2) only
D (1) and (3) only

★3 Ext 6.3 Two particles travel through a region with a uniform electric field and a uniform magnetic field without deflection. They have the same

A charge.
B mass.
C charge to mass ratio.
D velocity.

★4 6.3 *P*, *Q* and *R* are three long straight parallel wires carrying the same amount of current (Fig b). The currents in *P* and *R* flow into the page and the current in *Q* flows out of the page. *Q* is placed midway between *P* and *R*. The magnitude of the magnetic force per unit length acting on *R* by *P* is *F*. Find the resultant magnetic force per unit length acting on *R* by *P* and *Q*.



Fig b

A *F* (towards left)
B *F* (towards right)
C 3*F* (towards left)
D 3*F* (towards right)

★5 Ext 6.3 Some electrons undergo uniform circular motion of radius *r* in a magnetic field *B* (Fig c). The field is perpendicular to the direction of motion of the electrons. The charge and the mass of each electron are $-q$ and m respectively.

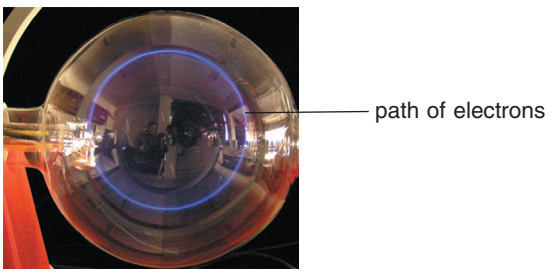


Fig c

Which of the following statements is/are correct?

(1) If the speed of the electrons is halved, the radius of the circular path will be 0.5*r*.

(2) The time that the electrons take to complete the path once is $\frac{2\pi m}{B}$.

(3) The electrons are accelerated by the magnetic field.

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

★6 6.3 Three long straight parallel wires X, Y and Z are located at the corners of a square (Fig d). Each wire carries the same current I. Find the direction of the resultant magnetic force acting on Z.

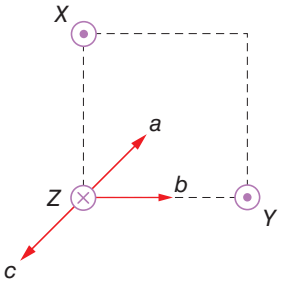


Fig d

A Direction a
B Direction b
C Direction c
D Zero resultant magnetic force

★7 6.1 A student constructs a battery car using a battery, a wire and two circular magnets (Fig e). The battery car moves to the left.

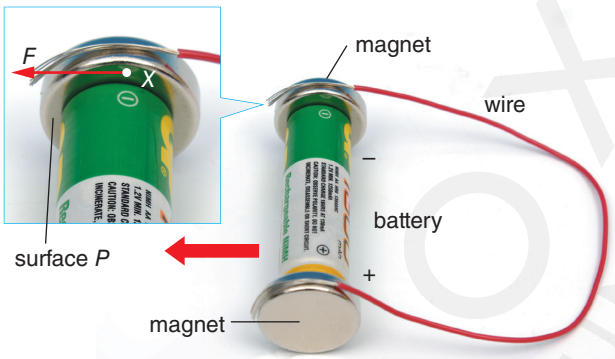


Fig e

X is a point on the magnet surface P, which is attached to the negative terminal of the battery. The magnetic force F acting on X points to the left, causing the magnet to rotate. Which of the following best describe the direction of the current I through X and the polarity of surface P?

	Direction of I	Polarity of surface P
A	upwards	N-pole
B	downwards	N-pole
C	upwards	S-pole
D	downwards	S-pole

→ This video demonstrates the battery car.

Video

★8 Ext 6.3 Figure f shows a charged particle P moving in a region with electric field E and magnetic field B mutually perpendicular to each other. At the instant shown, P is moving with a velocity v which is perpendicular to both fields. The dotted line shows its path afterwards.

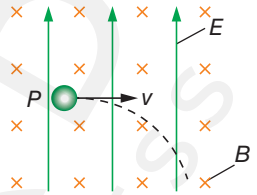


Fig f

Which of the following about P must be correct?

(1) P is negatively charged.

(2) The kinetic energy of P increases when it travels in the fields.

(3) $v \neq \frac{E}{B}$

A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

★★9 Ext 6.3 A charged particle moves in a straight line when passing through a uniform magnetic field between two parallel charged plates (Fig g).

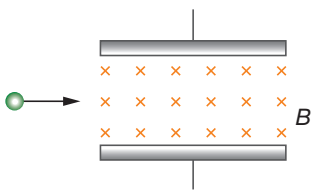


Fig g

Which of the following statements is/are correct?

(1) The particle must deflect upwards if it carries the opposite charge.

(2) The particle must deflect downwards if it moves at a lower speed.

(3) The upper plate must be the positive plate.

A (1) only
B (3) only
C (1) and (2) only
D (2) and (3) only

Refer p.250

★★ **10** 6.2 A student sets up a motor as shown (Fig h). Solenoids *X*, *Y* and coil *PQRS* are connected to the same power supply.

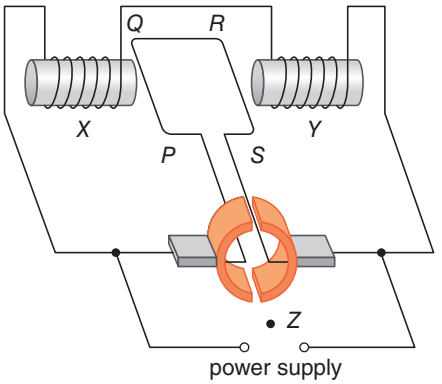


Fig h

Which of the following statements are correct?

(1) If a d.c. power supply is used, the coil will rotate clockwise when observed from point *Z*.

(2) If an a.c. power supply is used, the coil will only oscillate.

(3) The coil will rotate faster if *X* and *Y* have more turns per unit length.

A (1) and (2) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

Refer p.240

11 6.1 **HKDSE 2013 Paper 1A Q28**

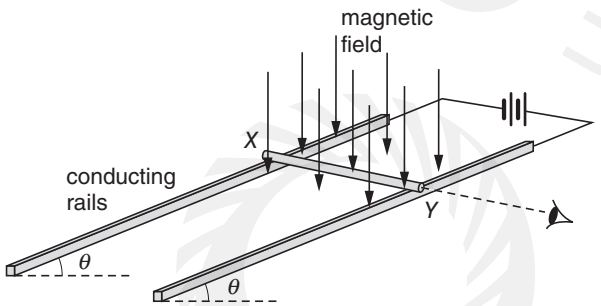
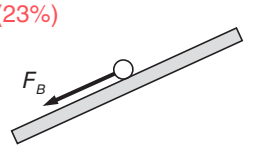


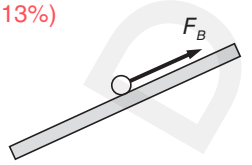
Fig i

A copper rod *XY* is placed on a pair of smooth inclined conducting rails which are located in a magnetic field applied vertically downwards. The rails make an angle θ to the horizontal and a battery is connected to the rails as shown (Fig i). Which diagram shown below represents the magnetic force F_B acting on the rod when viewed from end *Y*?

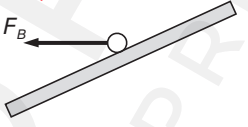
A (23%)



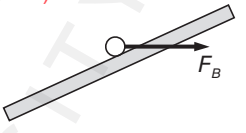
B (13%)



C (42%)



D (22%)



12 Ext 6.3 **HKDSE 2018 Paper 1A Q28**

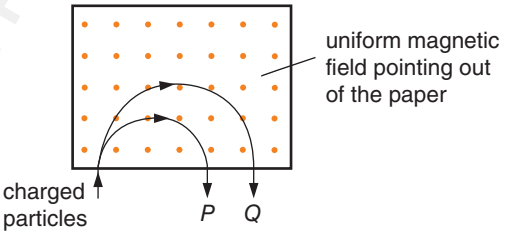


Fig j

P and *Q* are two particles carrying the same amount of charge but of different masses. They travel with the same speed and enter a uniform magnetic field pointing out of the paper as shown (Fig j). Semi-circular paths with different radii are described before they emerge from the field. Which descriptions below are correct?

(1) Both *P* and *Q* are positively charged.

(2) *P* and *Q* emerge from the field with the same speed.

(3) The mass of *Q* is greater than that of *P*.

A (1) and (2) only (19%)

B (1) and (3) only (28%)

C (2) and (3) only (17%)

D (1), (2) and (3) (36%)

13 HKDSE 2020 Paper 1A Q26
6.1

Figure k below shows the magnetic field pattern on a horizontal surface around a long vertical straight wire carrying a steady current I pointing out of the paper. The Earth's magnetic field is **not** neglected.

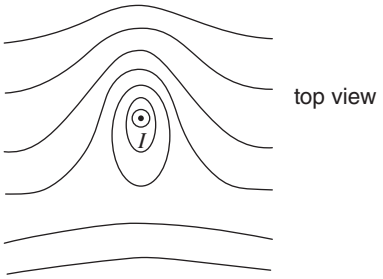


Fig k

What are the directions of the following?

	The horizontal component of the Earth's magnetic field	The magnetic force experienced by the current-carrying wire
A (42%)	←	↓
B (25%)	←	↑
C (13%)	→	↓
D (20%)	→	↑

Conventional questions

★**14** 6.1 Figure l shows a current-carrying wire between the poles of a magnet. The magnet is placed on a weighing scale.

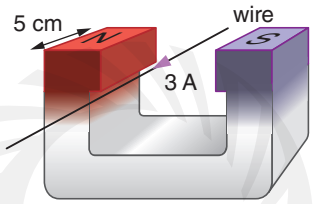


Fig l

(a) In Figure l, mark the direction of the magnetic force acting on the wire. (1 mark)

(b) The current in the wire is 3 A and the strength of the magnetic field produced between the poles is 0.04 T. If the scale reading shows 120 g, estimate the mass of the magnet. (3 marks)

(c) State the effect on the scale reading if

(i) the current is increased,

(ii) the current is reversed. (2 marks)

★**15** Ext 6.3 An electron moves inside a region with a magnetic field of 1.8 mT pointing into the page and an electric field produced by two parallel metal plates (Fig m). It keeps moving at a constant speed of 1.22×10^7 m s^{-1} without deflection. The mass and the charge of the electron are 9.11×10^{-31} kg and -1.60×10^{-19} C respectively.

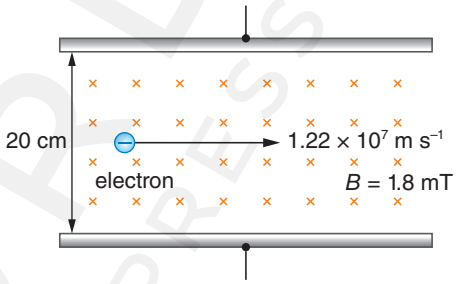


Fig m

(a) Find the magnitude and direction of the magnetic force acting on the electron. (3 marks)

(b) Find the strength and direction of the electric field. (3 marks)

(c) What is the voltage across the metal plates? (1 mark)

(d) (i) Describe the motion of a proton which enters this region horizontally with a speed higher than 1.22×10^7 m s^{-1}. (2 marks)

(ii) How should the voltage across the metal plates be adjusted to allow a charged particle with a speed higher than 1.22×10^7 m s^{-1} to pass through without deflection? (1 mark)

★**16** 6.3 Two long straight parallel wires P and Q are 1 cm apart (Fig n). Each wire carries a current of 2 A.

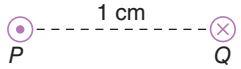


Fig n

When another long straight wire X is placed at 2 cm from Q, the resultant magnetic force acting on Q by P and X becomes zero. Find the possible position(s) of X and the corresponding size and direction of the current in it. (4 marks)

★17 6.1 A current balance consists of a movable rider of mass 5 g (Fig o). With no current, the frame is just balanced when the rider is 1 cm from *P*.

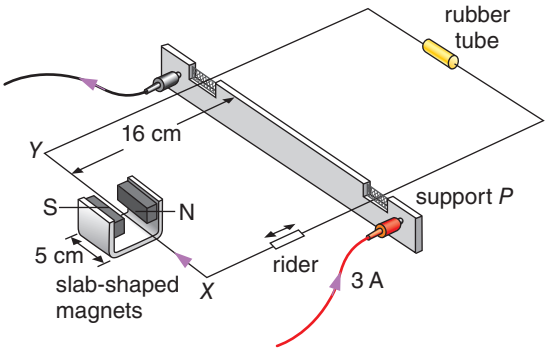


Fig o

Now, a current of 3 A flows in the *XY* direction. The rider is moved to a position 3.8 cm from *P* to retain the balance. Take the acceleration due to gravity $g = 9.81\ m\ s^{-2}$.

(a) Find the moment about *P* by the rider. (2 marks)

(b) Find the magnetic force acting on the frame. (2 marks)

(c) If the magnet is tilted at an angle such that the magnetic field is no longer perpendicular to *XY* (Fig p), how will the answer of (b) change? Show your calculation. (3 marks)

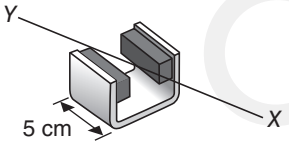


Fig p

(d) What will happen if the direction of the current is reversed? (1 mark)

★18 6.2 A student designs a simple d.c. motor as shown (Fig q). However, the motor does not work properly.

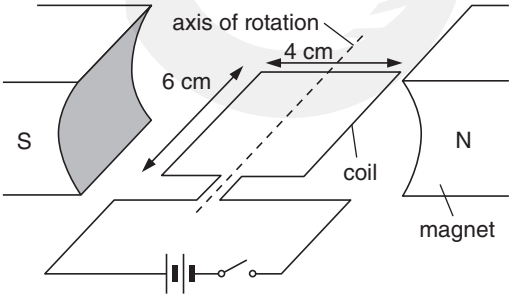


Fig q

(a) Describe the motion of the coil after the switch is closed. (2 marks)

(b) Suggest how he can improve his design so that the coil can rotate properly. (1 mark)

(c) The current flowing in the coil is 3 A. The magnetic field, which is assumed to be always parallel to the coil, has a magnitude of 0.02 T. Estimate the moment acting on the coil when the coil

(i) is at the horizontal position,

(ii) makes an angle of 15° with the horizontal.

Suppose the whole coil is within the magnetic field in both cases. (4 marks)

(d) Suggest two ways to increase the moment acting on the coil. (2 marks)

★19 6.1 Read the following passage about a **railgun** and answer the questions that follow.

A railgun makes use of electromagnetic force to launch a projectile (Fig r).

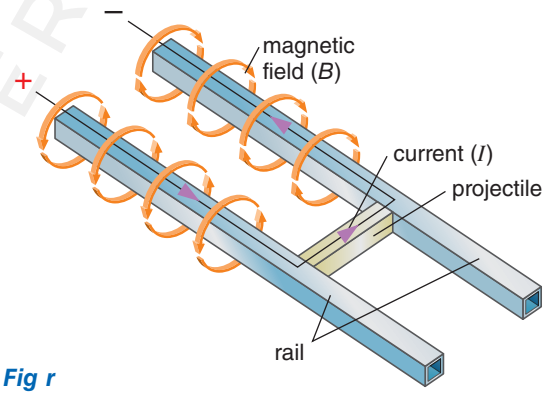


Fig r

It consists of a pair of conducting rails connected to a power supply. When a conducting projectile is placed on the rails, it completes the circuit. The magnetic field generated by the current in the rails produces a magnetic force on the current-carrying projectile and the projectile is launched.

(a) (i) Draw the direction of magnetic force acting on the projectile. (1 mark)

(ii) What is the answer in (i) if the current is reversed? Explain briefly. (2 marks)

(b) Suggest two methods to increase the speed of the projectile when it leaves the rails. (2 marks)

*20 6.2 The coil in a simple motor rotates clockwise (Fig s).

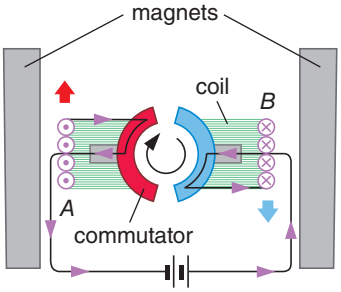


Fig s

(a) Mark the polarity of the magnets on Figure s. (1 mark)

(b) What is the function of the commutator? (2 marks)

(c) The magnetic forces do not produce turning effect on the coil at the instant the coil has turned 90°. How does the motor run continuously? (1 mark)

(d) Describe how the force acting on side *A* of the coil changes when the coil rotates. (2 marks)

(e) How will the turning effect by the magnetic forces change if the number of turns in the coil is increased? (1 mark)

*21 6.1 A current-carrying conducting rod of 5 cm long is placed in a uniform magnetic field of 4 mT (Fig t). The rod makes an angle of 30° with the magnetic field and experiences a magnetic force pointing into the page.

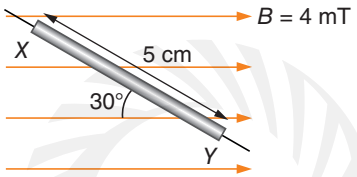


Fig t

(a) What is the direction of the current flowing in the rod? (1 mark)

(b) If the magnetic force acting on the rod is 1×10^{-3} N, find the current flowing in the rod. (2 marks)

(c) Describe the change in the magnitude and direction of the magnetic force acting on the rod if the rod is turned 90° clockwise on the plane of the page. (2 marks)

**22 Ext Misc A cyclotron is used in hospital to accelerate protons that are then targeted to kill cancer cells. The cyclotron consists of two D-shaped electrodes enclosed in a flat vacuum chamber, which is installed in the uniform magnetic field *B* between the two poles of a large magnet. Figure u shows the cyclotrons viewed from above. The magnet is not shown.

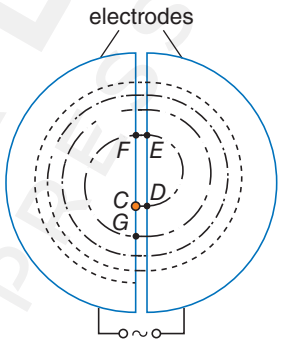


Fig u

A p.d. of 40 kV is set up across the narrow gap between the electrodes. The polarities of the electrodes reverse every time after the proton passes through the gap. Protons are released from rest at *C* and are accelerated across the gap by the p.d. The protons then travel a semi-circular path and are accelerated again across the gap. This process is repeated and the path of proton is shown in the diagram.

A proton has a charge of 1.60×10^{-19} C and a mass of 1.67×10^{-27} kg.

(a) (i) Find the work done on a proton every time it passes through the gap. Express your answer in J. (2 marks)

(ii) Find the speed of the proton as it reaches *D*. (2 marks)

(b) (i) What is the direction of the magnetic field in Figure u? (1 mark)

(ii) Show that the time taken for a proton to travel from *D* to *E* is $\frac{\pi m}{QB}$, where *m* and *Q* are the mass and charge of a proton. (3 marks)

(iii) Is the time taken for the proton to travel from *F* to *G* longer than, the same as, or shorter than the answer in (ii)? (1 mark)

(c) Does a cyclotron work with d.c. supply across the gap? Explain. (2 marks)

Refer p.249

★★ **23** 6.1 A student conducts an experiment to estimate the strength of the magnetic field between two magnets (Fig v). Magnets *P* and *Q*, each 7 cm long, on a steel yoke rest on an electronic balance. A wire with segment *XY* 5 cm long is suspended between them. By passing different currents from *X* to *Y*, different balance readings are obtained. The data recorded are shown in Table a.

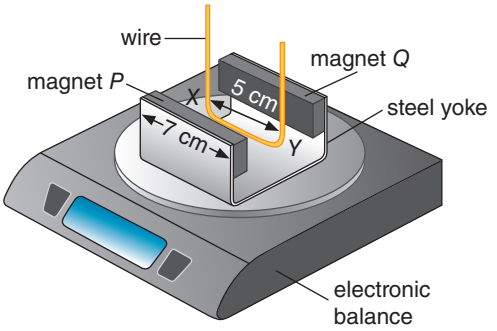


Fig v

Current / A	Balance reading / g
0	145.0
0.5	145.4
1.0	145.8
1.5	146.1
2.0	146.5

Table a

(a) Determine the direction of the magnetic force acting on the wire when current flows through the wire. (1 mark)

(b) Determine the pole of the inner face of *P*. (1 mark)

(c) Plot a suitable graph to estimate the strength of the magnetic field between *P* and *Q*. Take 1 g in the reading to be equivalent to 0.01 N. (4 marks)

▶ Refer p.228

See p.263 for the Exam Report.

24 6.1 **HKDSE 2015 Paper 1B Q9**

Figure w shows a set-up for demonstrating one of Faraday's discoveries. A light metal rod is free to rotate about point *P* while its lower end just touches some conducting liquid in a metallic container.

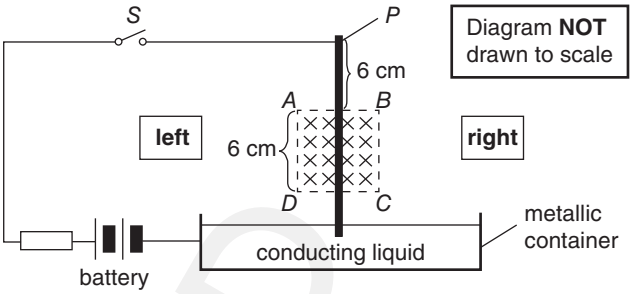


Fig w

A uniform magnetic field pointing into the paper is applied over the region *ABCD* containing part of the rod. When switch *S* is closed, the rod 'kicks' out and leaves the liquid surface.

(a) State the direction (to the left / to the right / into the paper / out of the paper) that the rod 'kicks' and describe the subsequent motion of the rod. (3 marks)

(b) When switch *S* is closed, the initial moment about point *P* that makes the rod 'kick' out is 7.2×10^{-4} N m. Assume that the magnetic force acts at the **midpoint of the part of the rod within the magnetic field**.

(i) Calculate the magnetic force acting on the rod at that instant. (2 marks)

(ii) Hence, find the strength *B* of the magnetic field if the current flowing through the rod is 3.2 A when the circuit is closed. (2 marks)

(c)

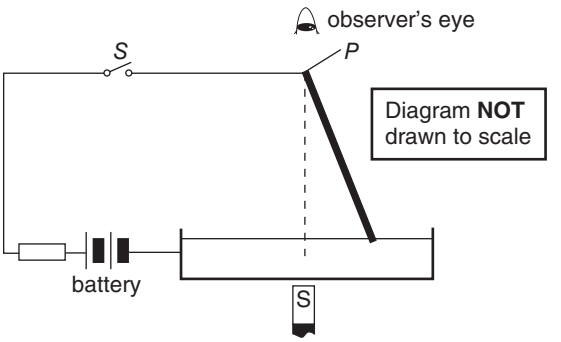


Fig x

Now the uniform magnetic field is removed and a bar magnet is placed underneath the container as shown in Figure x. The rod is held tilted at an angle to the vertical but with its lower end still in the conducting liquid.

(i) Sketch on Figure x the field lines around the rod due to the bar magnet. (1 mark)

(ii) After closing switch S and the rod is released from rest, describe its subsequent motion viewed from above. (1 mark)

25 HKDSE 2017 Paper 1B Q9

6.3

(a) Two long straight current-carrying wires, P and Q, are parallel to each other and lie on the plane of the paper as shown in Figure y. The currents in the wires, I_P and I_Q, flow in the same direction.

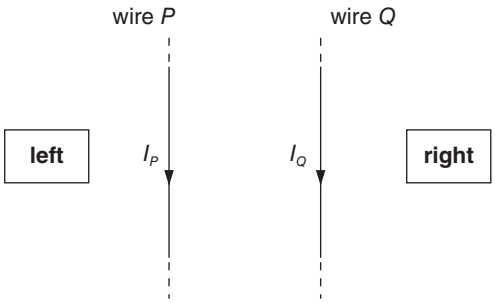


Fig y

(i) State the direction (to the left / to the right / into the paper / out of the paper) of the magnetic field at Q due to P. (1 mark)

(ii) In Figure y, draw the direction of the magnetic force acting on Q due to P. (1 mark)

(iii) Show that the magnitude of the magnetic force per unit length F_l, acting on Q due to P is $F_l = \dfrac{\mu_0 I_P I_Q}{2\pi r}$

where μ_0 is the permeability of free space and r is the separation between the two wires. (3 marks)

(iv) For the magnetic force acting on Q due to P and the magnetic force acting on P due to Q, if $I_P \neq I_Q$, briefly explain whether the two forces are equal in magnitude. (2 marks)

(b) Figure z shows a metal slinky spring.

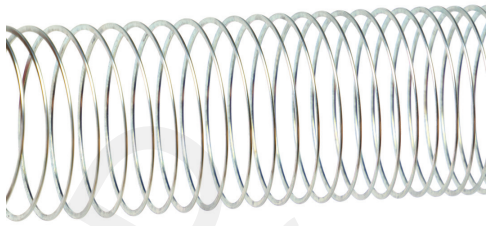


Fig z

(i) If a direct current passes through the spring, briefly explain whether the spring will be compressed or stretched due to magnetic force. (2 marks)

(ii) A student suggests that the spring will be compressed and stretched alternately due to magnetic force when an alternating current passes through. Briefly explain why he is wrong. (1 mark)

Q24 Exam Report: In (a), some candidates tried to explain the direction of the electromagnetic force acting on the rod rather than to describe its subsequent motion. In (b)(i), most were able to apply the definition of moment ($F \times d$) to solve this problem but many of them failed to identify the correct value of d and ended up with an incorrect numerical answer. Some candidates did not convert cm into m in their calculation. Incorrect equation (e.g. $B = \dfrac{\mu_0 I}{2\pi r}$) or unit (e.g. Bq, Wb) were quoted in (b)(ii). In (c)(i), some candidates treated the rod as a bar magnet and sketched the field pattern wrongly. Common errors included incorrect direction of field lines, none of the field lines crossed the rod or uniform field patterns. Not many were able to state that the rod performs circular motion in an anti-clockwise direction in (c)(ii).

Q25 Exam Report: This question tested candidates' knowledge and understanding on electromagnetism. Their performance was fair. Part (a) was in general well answered except (a)(iv) in which not many were able to explain why the two magnetic forces, a pair of action and reaction forces, were equal in magnitude. In (b)(i), only the more able candidates referred to the direction of currents in neighbouring wire segments when explaining the origin of magnetic forces. Some misunderstood the situation in (b)(ii) and tried to answer the question by considering the induced current in the spring.

Know more

Moving-coil loudspeakers (p.223)

A moving-coil loudspeaker (Fig a) contains a short free coil inside a cylindrical magnet, where the wire is at right angles to the field. As an alternating current flows forwards and backwards through the coil, the coil is pushed in and out (Fig b). This makes the cone vibrate to emit sound waves.



Fig a

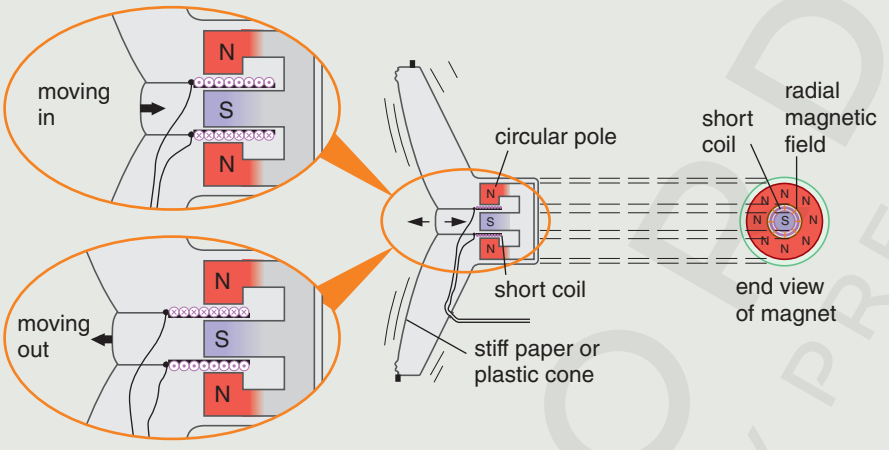


Fig b

→ This simulation demonstrates how a loudspeaker works. It is available on OUP web in PC version (.exe) only.

Simulation

Battery train (p.223)

You can make a battery train using some magnets and a battery (Fig c).

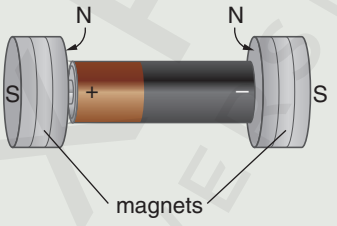


Fig c

When the train is put inside a copper coil (Fig d), the magnets, which are made of metal, touch points *P* and *Q* of the coil and form a closed circuit. Figure e shows the direction of the current in the coil.

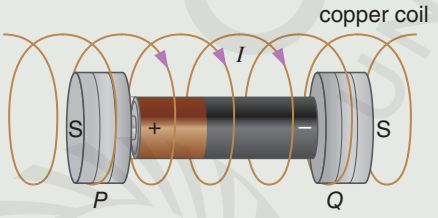


Fig d

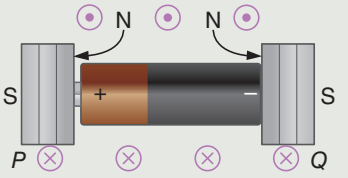


Fig e

Figure f shows the magnetic field between the magnets. By Fleming's left-hand rule, the magnetic force acting on the coil points left.

Stress that Fleming's left-hand rule can be used to find the direction of the magnetic force experienced by a current-carrying conductor (the coil, not the magnet, as in this case).

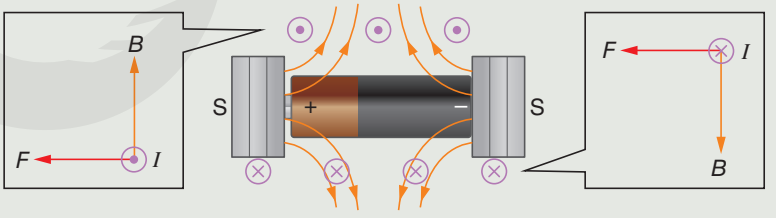


Fig f

By Newton's third law, when the magnets exert a force on the coil, the coil also exerts a force on the magnets (the train) in the opposite direction. Therefore, the train moves to the right.

Know more

Deriving $F = BQv$ from $F = BIl$ (p.245)

Consider a conductor of length l carrying current I in a direction perpendicular to a uniform magnetic field B. The magnetic force acting on the conductor F_1 is given by:

$$F_1 = BIl$$

F_1 can be considered as the sum of the magnetic force acting on each moving charge. The force on each charge F_2 is given by:

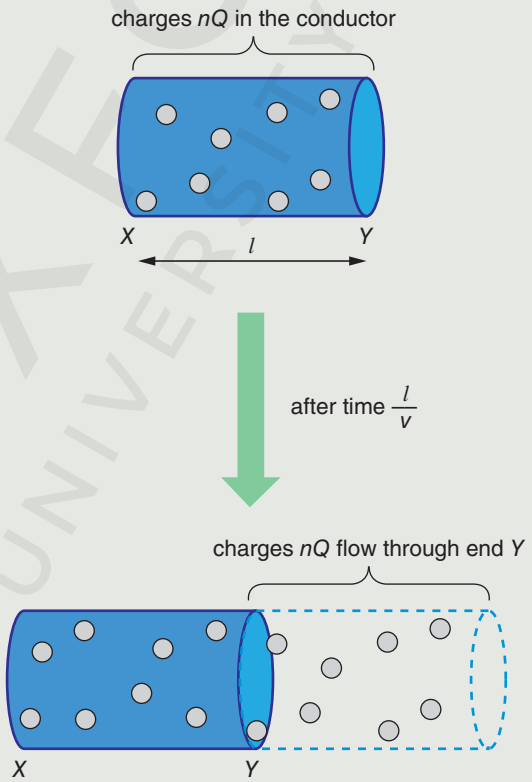
$$F_2 = \frac{BIl}{n}$$

where n is the number of charges in the conductor.

Assume each charge moves along the conductor with an average velocity v and each carries charge Q.

The average time taken for a charge to move from end X to end Y is given by $\frac{l}{v}$.

During this time period, the same quantity of charge as that originally in the conductor flows through the cross-section of end Y.



According to the definition of current,

$I = \frac{\text{charge}}{\text{time}}$ ▶

$$I = nQ \div \frac{l}{v} = \frac{nQv}{l}$$

Therefore,

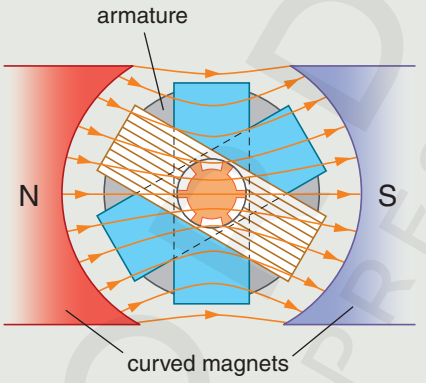
$$F_2 = \frac{Bl}{n} \times \frac{nQv}{l} = BQv$$

Know more

Practical motors (p.239)

The soft-iron core can strengthen the magnetic field and increase the turning effect. ▶

A practical motor usually uses a soft-iron core with several coils, which is called an *armature*. Moreover, a curved magnet is used instead of a flat one so that the turning effect can be made more uniformly when the coil turns.





Some practical motors make use of electromagnets rather than permanent magnets, so that they can run on a.c.

armature 電樞